Computer-Aided VLSI System Design



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國內外高階人才計畫

數位晶片設計實務演練



經濟部產業發展署
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Ministry of Economic Affairs



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INSTITUTE FOR INFORMATION INDUSTRY

Graduate Institute of Electronics Engineering, National Taiwan University

Advanced Computer-Aided VLSI System Design

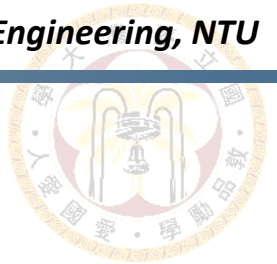
Chapter 7. Cell-Based APR Flow Part1

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Graduate Institute of Electronics Engineering, National Taiwan University



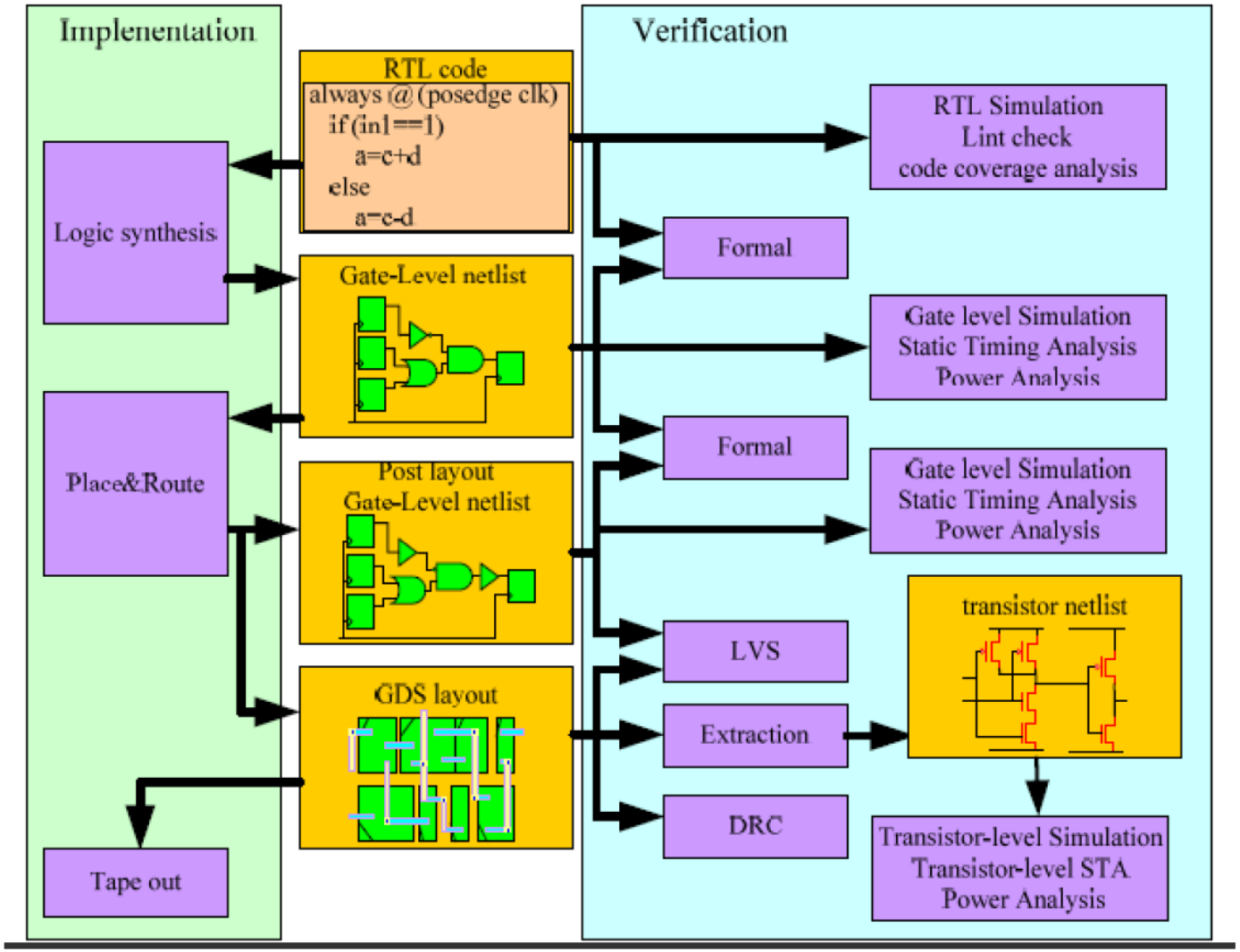
NTU GIEE



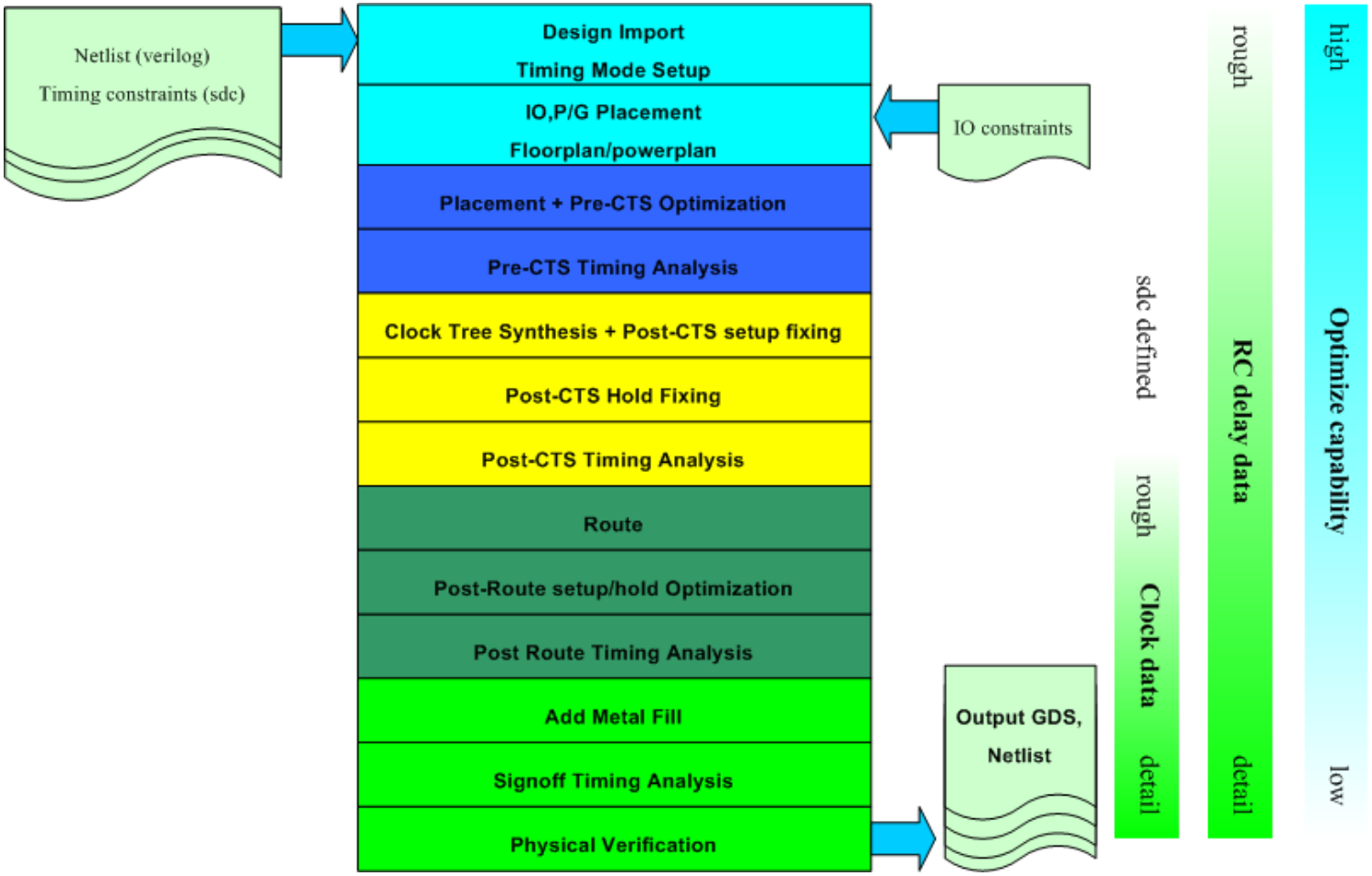
Outline

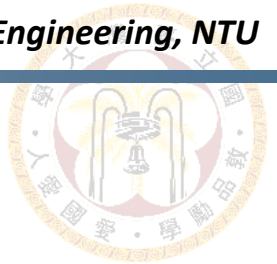
- Introduction
- Data Preparation
- Automatic Place & Route – Innovus
  - Design Import
  - Floorplan Specification
  - Power Planning

Cell-Based Design Flow



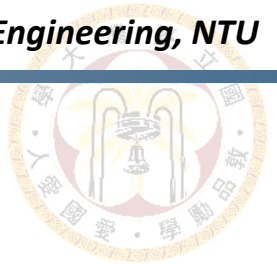
Innovus APR Flow





Outline

- Introduction
- **Data Preparation**
- Automatic Place & Route – Innovus
  - Design Import
  - Floorplan Specification
  - Power Planning



Data Preparation

- Library
  - Physical library (LEF: technology, abstract)
  - Timing library (LIB)
  - Parasitic File
- User Data
  - Gate-level netlist
  - SDC constraint
  - MMMC file
  - IO constraint
  - Scan def

Data Preparation: Physical Library

- Library Exchange Format (LEF)
  - Abstract physical view of standard cells and macros
  - Cell dimensions, pins, metal layers…
  - Simplified the layout for APR flow
  - Text file: human readable

- Graphic Design System (GDS)
  - Detailed geometric of standard cells and macros
  - Exact shapes, layers, text…
  - Stream out GDS for final tape-out
  - Binary file

Example LEF File of Standard Cell

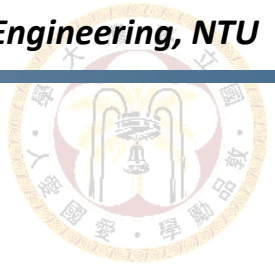
```
MACRO AN2D12BWP16P90
    CLASS CORE ;
    FOREIGN AN2D12BWP16P90 0.000 0.000  ;
    ORIGIN 0.000 0.000 ;
    SIZE 1.800 BY 0.576 ;
    SYMMETRY X Y ;
    SITE core ;
    PIN Z
        ANTENNADIFFAREA 0.001 ;
        DIRECTION OUTPUT ;
        USE SIGNAL ;
        PORT
        LAYER M1 ;
        RECT  0.688 0.132 0.754 0.228 ;
        RECT  0.688 0.348 0.754 0.444 ;
        RECT  0.868 0.132 0.934 0.228 ;
```

- Cell name
- Standard cell
- Size
- Can X/Y-axis flip
- Output pin
- M1 shape

Data Preparation: LIB

- Different operation conditions have independent .lib
  - Corner: slow, fast, typical
  - Voltage: nominal voltage, nominal voltage ± 10%
  - Temperature: -40°C, 25°C, 125°C
- Timing constraint
  - Setup, hold, recovery, removal…
- Power information
- Noise information
  - CCS/ECSM/cdB include, NLDM does not

Timing Model
NLDM: non-linear delay model (Traditional)
CCS: composite current source
ECSM: effective current source model

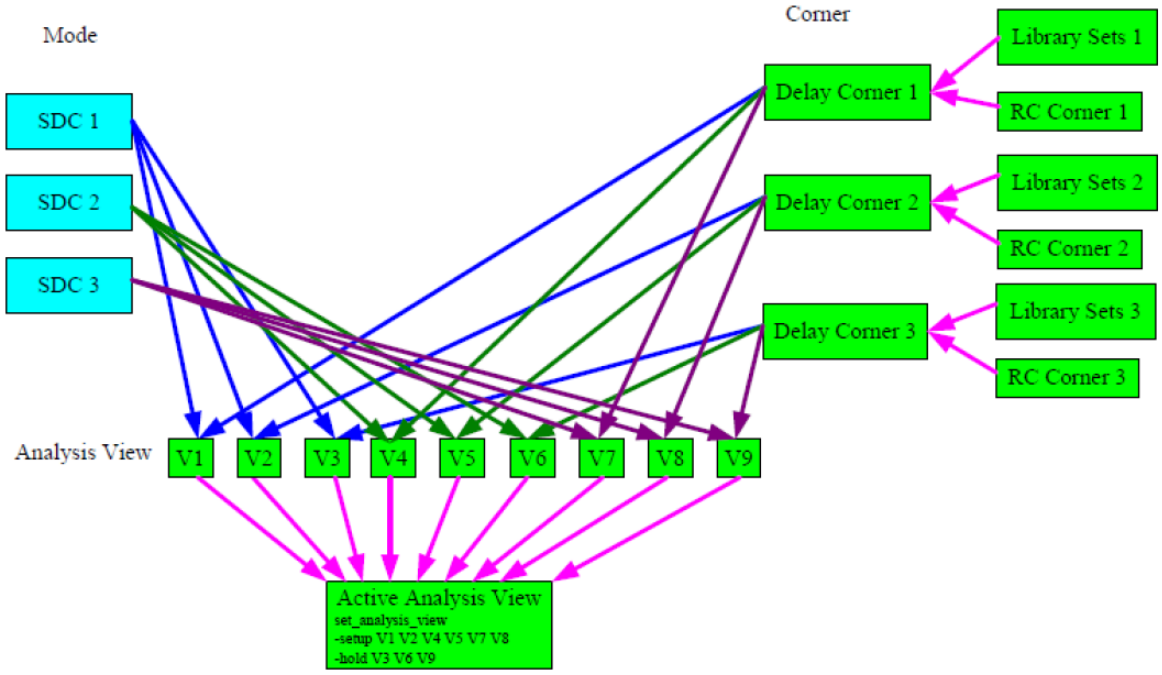


Data Preparation: Parasitic File

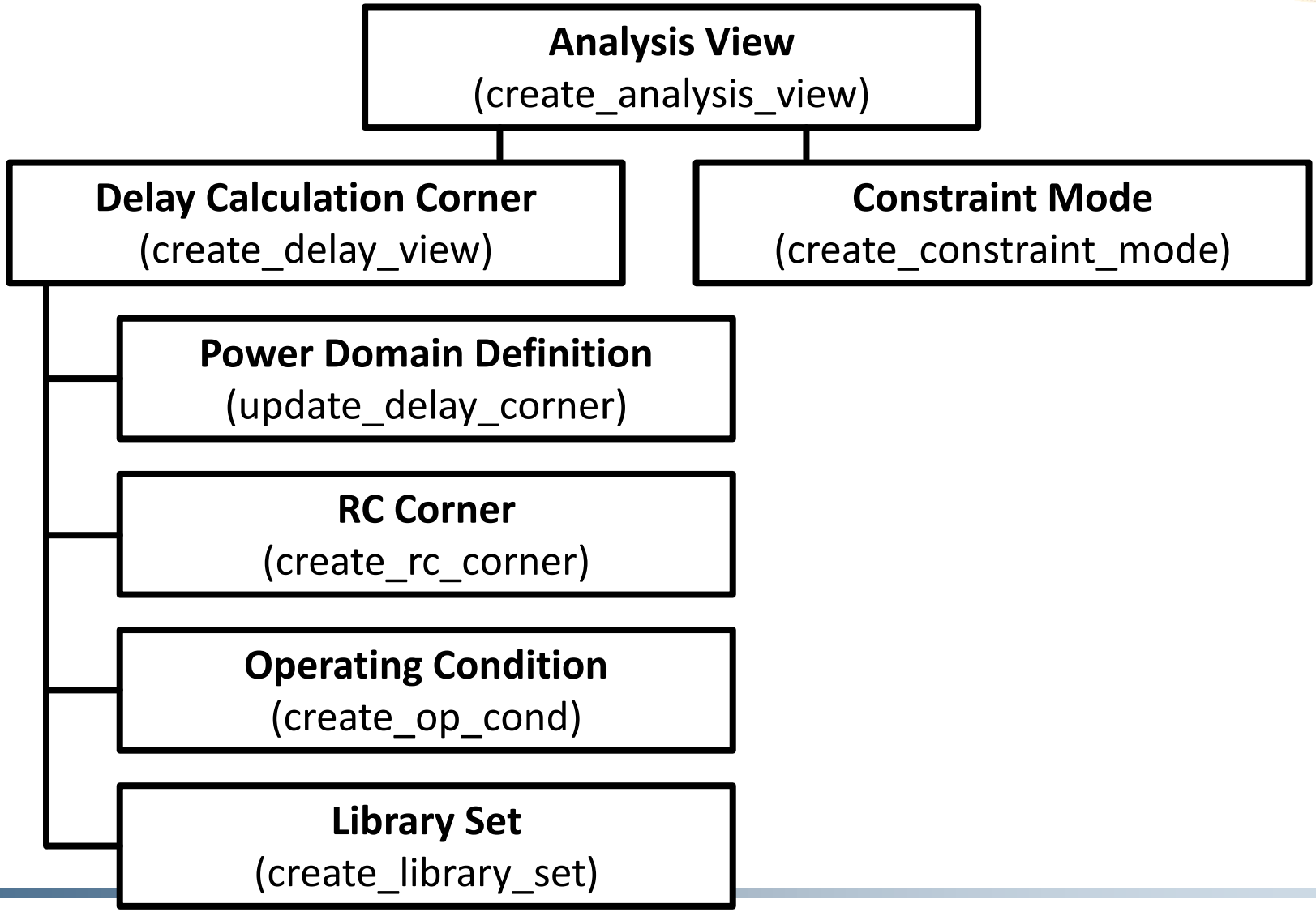
- Interconnect Technology Format (ICT or ITF)
  - Resistivity, dielectric constant, temperature coefficient…
  - Do not use these files directly for RC extraction
- 2D models (Capacitance Table or TLU+)
  - Simplified models for RC extraction
  - Lookup table of width, spacing… to get RC
- 3D models (qrcTechfile or nxtgrd)
  - More accurate model than 2D models
- Can use 2D model or 3D model for APR

Data Preparation: MMMC View

- Example: Design is required to meet 3 operating corners
  - Corner1: 0.72V, -40°C
  - Corner2: 0.8V, 25°C
  - Corner3: 0.88V, 125°C



MMMC Structure



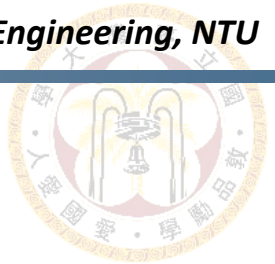
Example of MMMC

- Max view: max delay + worst RC
- Min view: min delay + best RC
- Need to consider high and low temperature in both max and min delay

```
create_rc_corner -name RC_worst -preRoute_res {1.0} -preRoute_cap {1.0} -preRoute_clkres {0.0} -
preRoute_clkcap {0.0} -postRoute_res {1.0} -postRoute_cap {1.0} -postRoute_xcap {1.0} -postRoute_clkres {0.0} -
postRoute_clkcap {0.0} -qx_tech_file {worst/qrcTechFile}
create_rc_corner -name RC_best -preRoute_res {1.0} -preRoute_cap {1.0} -preRoute_clkres {0.0} -
preRoute_clkcap {0.0} -postRoute_res {1.0} -postRoute_cap {1.0} -postRoute_xcap {1.0} -postRoute_clkres {0.0} -
postRoute_clkcap {0.0} -qx_tech_file {best/qrcTechFile}

create_library_set -name lib_0p72v_m40c -timing {StdCellss0p72vm40c_ccs.lib StdIOss0p72v1p62vm40c.lib
Memoryss_0p72v_0p72v_125c_syn.lib}
create_library_set -name lib_0p88v_m40c -timing {StdCellff0p88vm40c_ccs.lib StdIOff0p88v1p98vm40c.lib
Memoryff_0p88v_0p88v_m40c_syn.lib}
create_library_set -name lib_0p72v_125c -timing {StdCellss0p72v125c_ccs.lib StdIOss0p72v1p62v125c.lib
Memoryss_0p72v_0p72v_125c_syn.lib}
create_library_set -name lib_0p88v_125c -timing {StdCellff0p88v125c_ccs.lib StdIOff0p88v1p98v125c.lib
Memoryff_0p88v_0p88v_m40c_syn.lib}

create_constraint_mode -name CM_func -sdc_files {CHIP_func.sdc}
create_constraint_mode -name CM_scan -sdc_files {CHIP_scan.sdc}
```



Example of MMMC

- Setup: max view
- Hold: take whole view into consideration
  - Max library also has longer hold time requirement

```
create_delay_corner -name DC_0p72v_m40c -library_set {lib_0p72v_m40c} -rc_corner {RC_worst}
create_delay_corner -name DC_0p72v_125c -library_set {lib_0p72v_125c} -rc_corner {RC_worst}
create_delay_corner -name DC_0p88v_m40c -library_set {lib_0p88v_m40c} -rc_corner {RC_best}
create_delay_corner -name DC_0p88v_125c -library_set {lib_0p88v_125c} -rc_corner {RC_best}

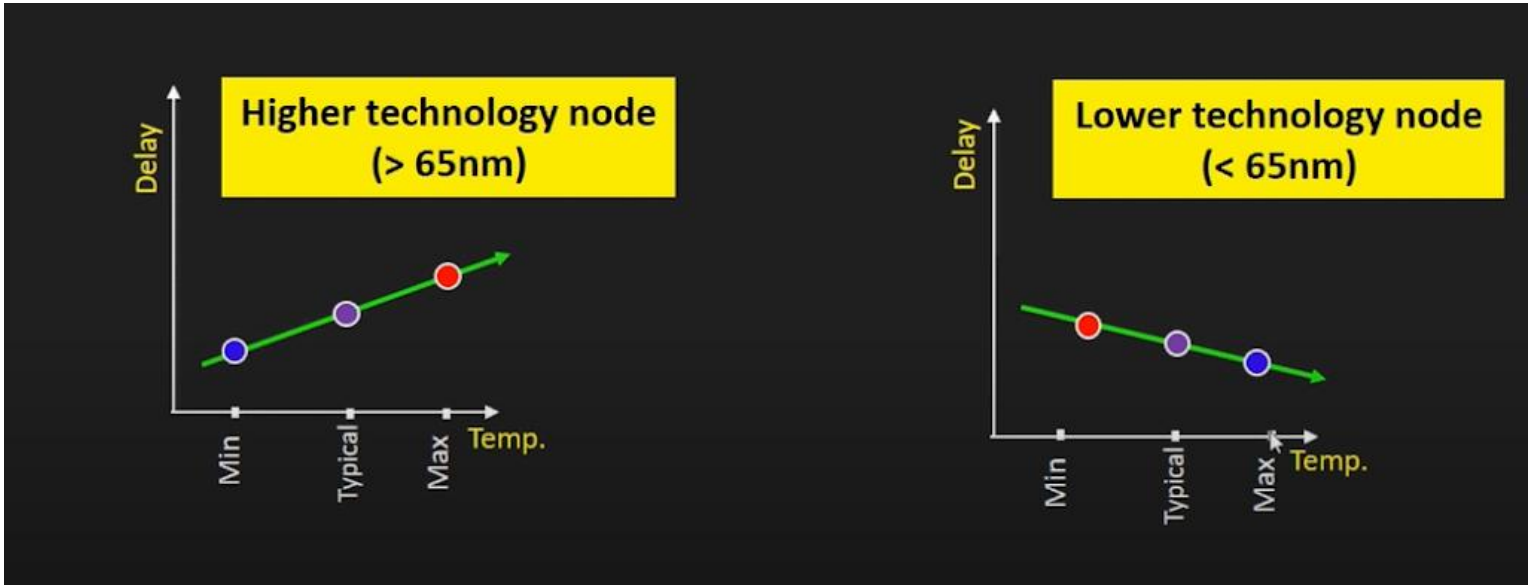
create_analysis_view -name AV_func_0p72v_m40c -constraint_mode {CM_func} -delay_corner {DC_0p72v_m40c}
create_analysis_view -name AV_func_0p72v_125c -constraint_mode {CM_func} -delay_corner {DC_0p72v_125c}
create_analysis_view -name AV_func_0p88v_m40c -constraint_mode {CM_func} -delay_corner {DC_0p88v_m40c}
create_analysis_view -name AV_func_0p88v_125c -constraint_mode {CM_func} -delay_corner {DC_0p88v_125c}
create_analysis_view -name AV_scan_0p72v_m40c -constraint_mode {CM_scan} -delay_corner {DC_0p72v_m40c}
create_analysis_view -name AV_scan_0p72v_125c -constraint_mode {CM_scan} -delay_corner {DC_0p72v_125c}
create_analysis_view -name AV_scan_0p88v_m40c -constraint_mode {CM_scan} -delay_corner {DC_0p88v_m40c}
create_analysis_view -name AV_scan_0p88v_125c -constraint_mode {CM_scan} -delay_corner {DC_0p88v_125c}

set_analysis_view -setup {AV_func_0p72v_m40c AV_func_0p72v_125c AV_scan_0p72v_m40c
AV_scan_0p72v_125c} -hold {AV_func_0p72v_m40c AV_func_0p72v_125c AV_func_0p88v_m40c
AV_func_0p88v_125c AV_scan_0p72v_m40c AV_scan_0p72v_125c AV_scan_0p88v_m40c AV_scan_0p88v_125c}
```

Temperature Inversion

- In advanced technology, cell delay is lower in higher temperature
- temperature ↑, mobility ↓, V_{th} ↓
- Since V_{gs} is closer to V_{th} in advanced technology, I_d is more sensitive to V_{th}

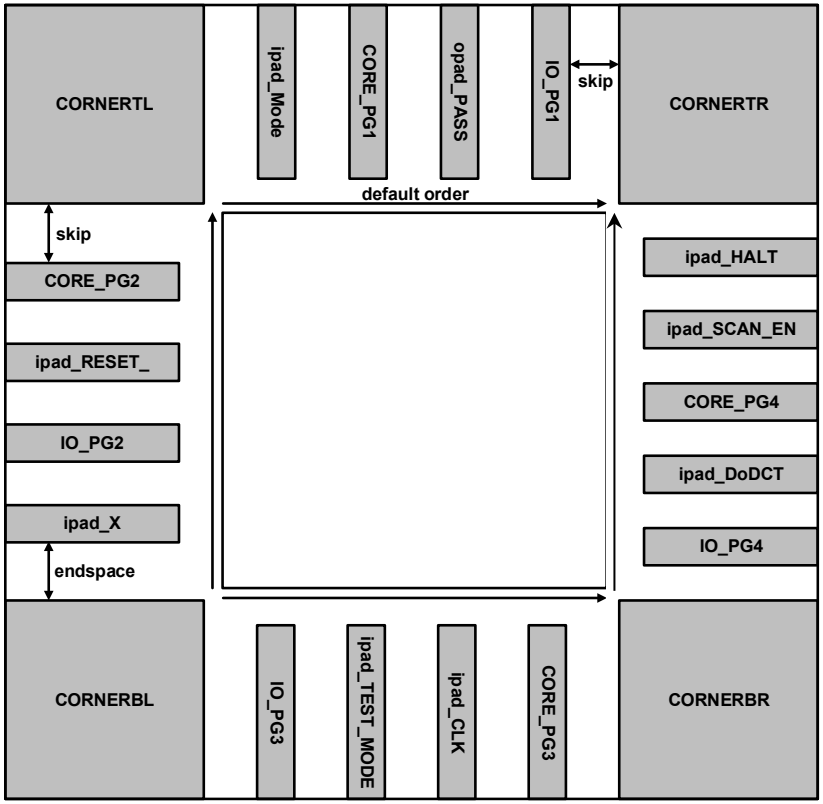
$$I_d = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left(V_{gs} - V_{th}\right)^{\alpha}$$



Source: https://teamvlsi.com/2021/05/temperature-inversion-in-vlsi.html#google_vignette

Data Preparation: I/O Assignment File

```
(globals
   version = 3
   io_order = default
)
(iopad
   (topright
   (inst name="CORNERTR" cell="PCORNER")
   )
   (top
   (inst  name="ipad_Mode" skip=60)
   (inst  name="CORE_PG1" cell="PVDD1CDGM_V")
   (inst  name="opad_PASS")
   (inst  name="IO_PG1" cell="PVDD2CDGM_V")
      (endspace gap=60.48)
   )
   (topleft
   (inst name="CORNERTL" cell="PCORNER" orientation=MX)
   )
   (left
   (inst  name="ipad_X" skip=60)
   (inst  name="IO_PG2" orientation=MX90)
   (inst  name="ipad_RESET_")
   (inst  name="CORE_PG2" cell="PVDD1CDGM_H" orientation=MX90)
      (endspace gap=60.48)
   )
   …
)
```



Data Preparation: I/O Assignment File

```
(globals
   version = 3
   io_order = default
)
(iopad
   (topright
   (inst name="CORNERTR" cell="PCORNER")
   )
   (top
   (inst  name="ipad_Mode" skip=60)
   (inst  name="CORE_PG1" cell="PVDD1CDGM_V")
   (inst  name="opad_PASS")
   (inst  name="IO_PG1" cell="PVDD2CDGM_H")
      (endspace gap=60.48)
   )
   (topleft
   (inst name="CORNERTL" cell="PCORNER" orientation=MX
   )
   (left
   (inst  name="ipad_X" skip=60)
   (inst  name="IO_PG2" orientation=MX90)
   (inst  name="ipad_RESET_")
   (inst  name="CORE_PG2" cell="PVDD1CDGM_H" orientation=MX90)
      (endspace gap=60.48)
   )
   …
)
```

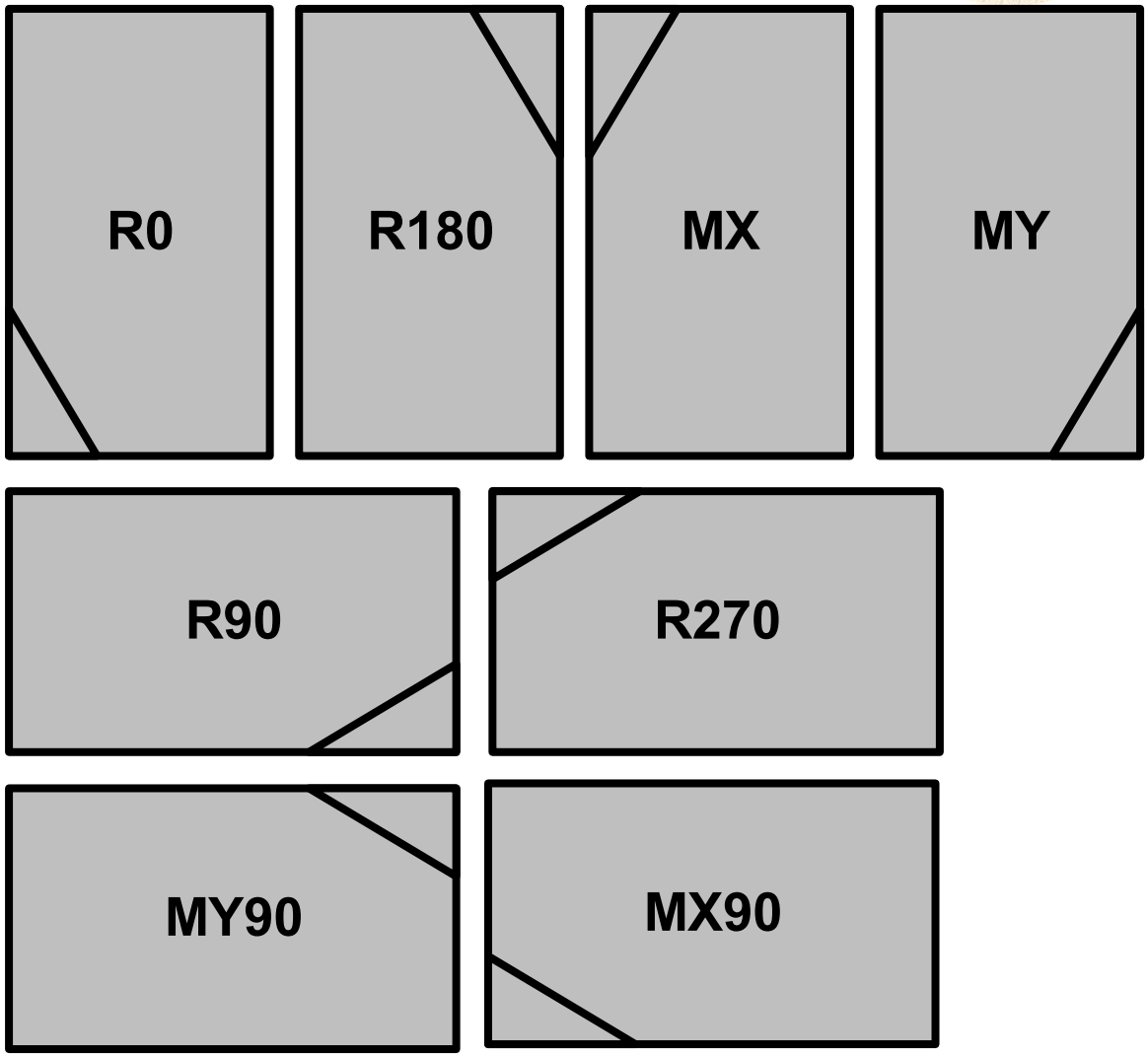
Order of I/O pins: clockwise, counterclockwise, default

If netlist does not include power pad, assign here

R0, R90, R1820, R270, MX, MY, MX90, MY90

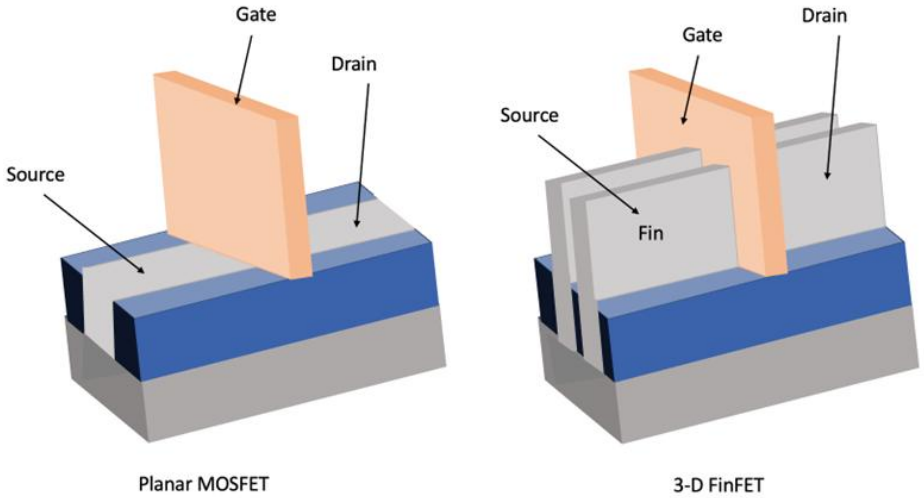
Orientation

- Eight types of orientation
- May need to adjust orientation of IO pads or macros
  - Power rails of pads
  - Uni-poly direction



Comparison between Different Processes

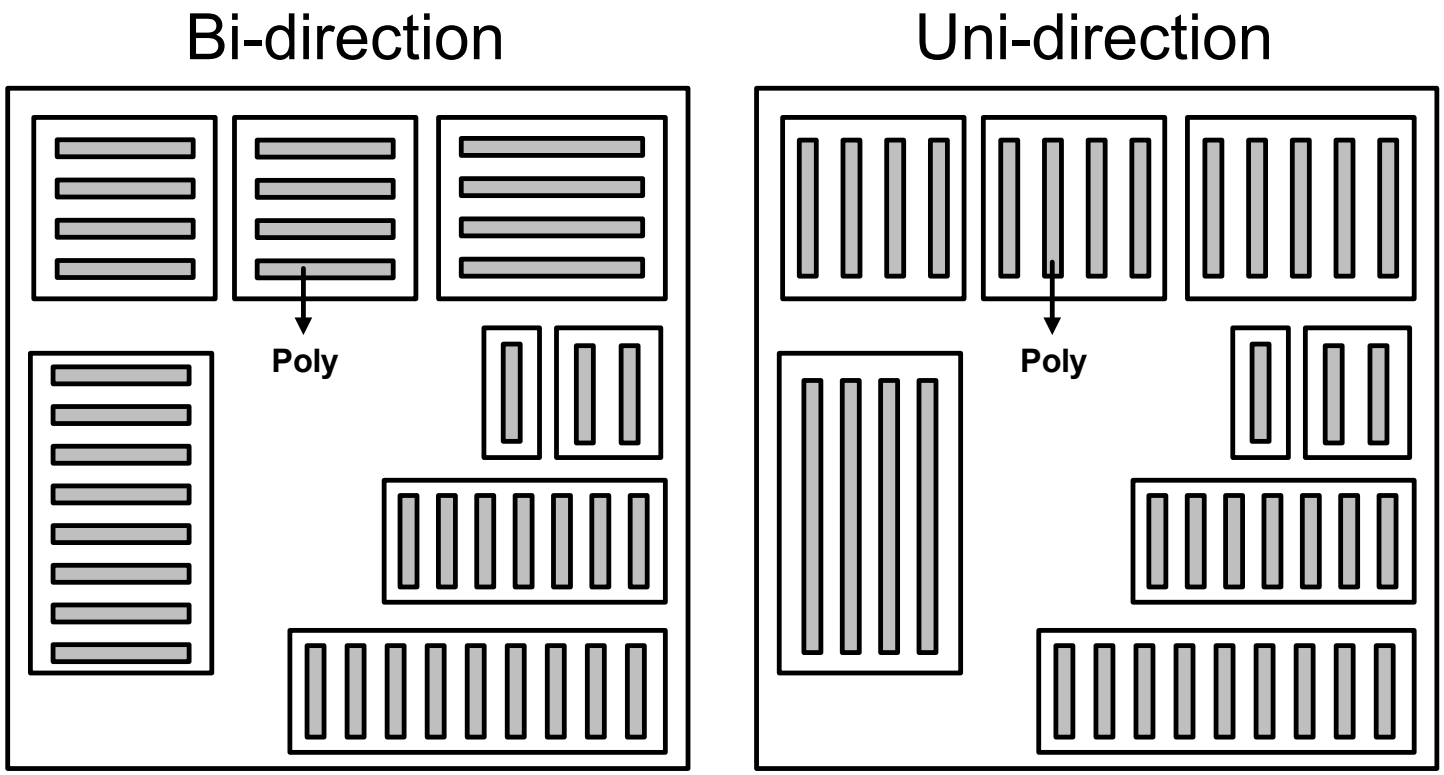
	90nm	N16 ADFP
Structure	Planar	FinFET
Poly Direction	Bi-direction	Uni-direction
Placement Grid	No	Yes
Boundary Cell	No	Yes
Well Tap	No	Yes



Source: https://anysilicon.com

Uni-Direction of Poly

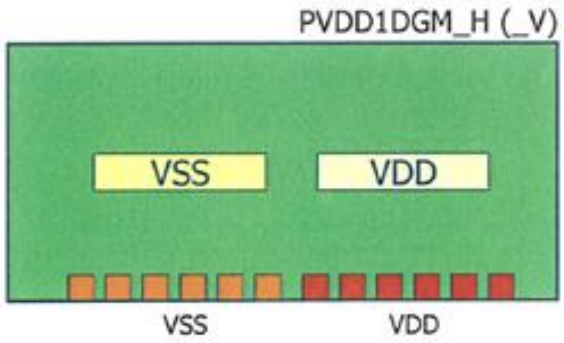
- Poly must be vertical in N16 ADFP
  - Restrict cell orientation including IO pads, macros,…

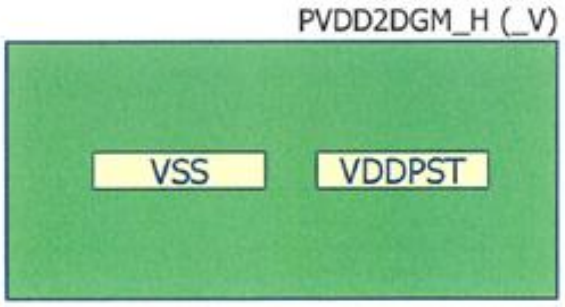


Power Pads of ADFP

- Both VDD and VSS in a single pad in N16 ADFP

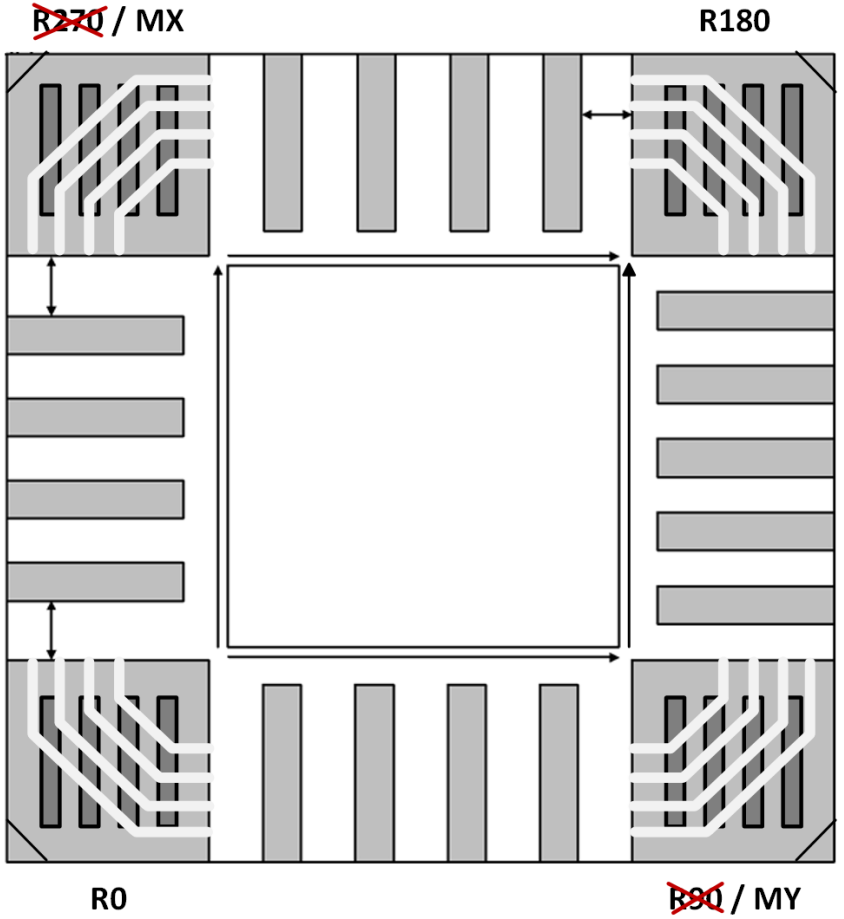
Cell Name	Features
PVDD1CDGM_H (_V)	Digital Core Power + Ground Combo Pad
PVDD2CDGM_H (_V)	Digital IO Power Pad
PFILLER00001	Digital Filler
PCORNER	Digital Corner





IO Pads Planning in ADFP

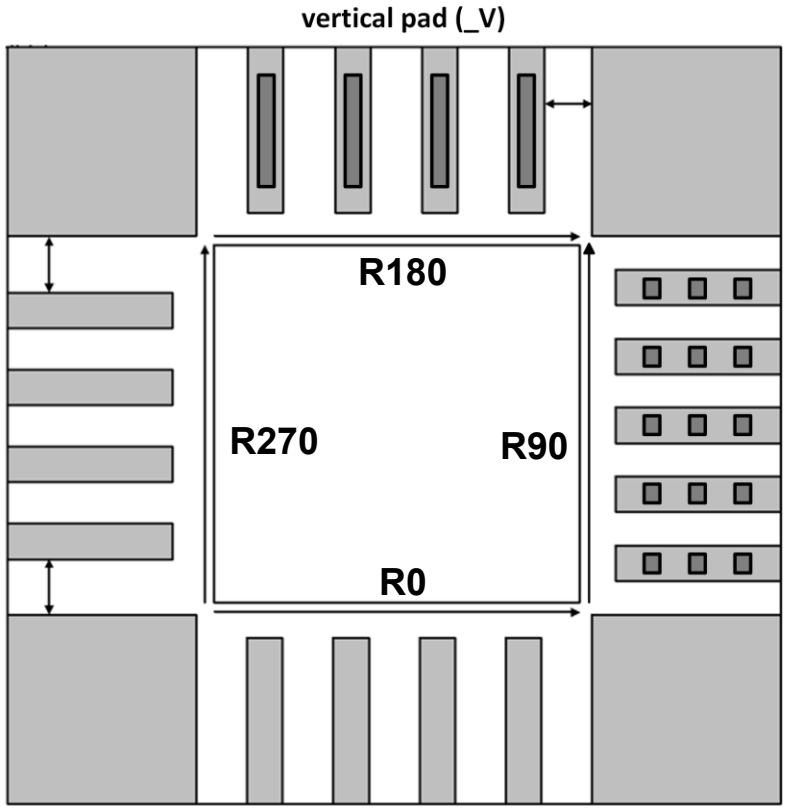
- ADFP corner cell is named PCORNER
  - Bottom left orientation
    - Innovus default: R0
  - Bottom right orientation
    - Innovus default: ~~R90~~
    - MY is correct
  - Top right orientation
    - Innovus default: R180
  - Top left orientation
    - Innovus default: ~~R270~~
    - MX is correct

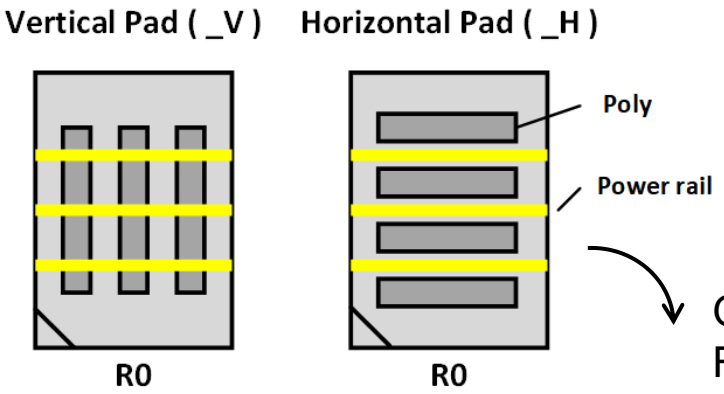


IO Pads Planning in ADFP

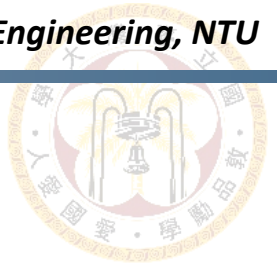
- Left and right: use _H pads
- Top and bottom: use _V pad
- May use wrong type of pads in netlist since you don't know the IO pads' positions during synthesis
  - Innovus change cell command

```
ecoChangeCell -inst $original_cell -cell $update_cell
```





Orientation of horizontal pads should be R90, R270, MX90, or MY90



How to Choose Signal Pads?

- Use the same type of input signal pads in most cases
- Choose output signal pads according to the output load
  - Larger output load: use pads that can provide larger current
    - Pros: lower transition time, can achieve higher frequency
    - Cons: higher power, need more IO power pads
- Only one kind of signal pad in ADFP: PDCDG_H (_V)
  - Left and right: use _H pads
  - Top and bottom: use _V pad

How Many IO Power Pads Do We Need?

- Simultaneously switching outputs (SSO)
- Driving index (DI)
  - maximum number of copies of an I/O cell switching from high to low simultaneously without making the voltage on the quiet output “0” higher than threshold value “Vil” when a single ground cell is applied
- Driving factor (DF): DF = 1/DI
- Sum of driving factor (SDF)
- SSO case
  - Number of IO VDD pads: SDF/1.1
  - Number of IO VSS pads: SDF/1.0
- Non-SSO case (most cases)
  - Number of IO VDD pads: SDF/1.6
  - Number of IO VSS pads: SDF/1.5

# Ground Pad	# Output Pad
1	DI
DF	1

Example of Determining # of IO Power Pads

- IO power pads are used to provide power for signal pads
- Only need to consider output signal pads when determining the number of IO power pads
- If a chip has 20 PDD02DGZ (2 mA), 10 PDD16DGZ (16 mA) output signal pads
  - Sum of DF = 20 * 0.02 + 10 * 0.3 = 3.4
  - In Non-SSO case
    - Number of IO VDD pad = 3.4 / 1.6 = 2.125 -> 3
    - Number of IO VSS pad = 3.4 / 1.5 = 2.267 -> 3

IO Type	2 mA	4 mA	8 mA	12 mA	16 mA	24 mA
DF Value	0.02	0.03	0.09	0.18	0.3	0.56

How Many Core Power Pads Do We Need?

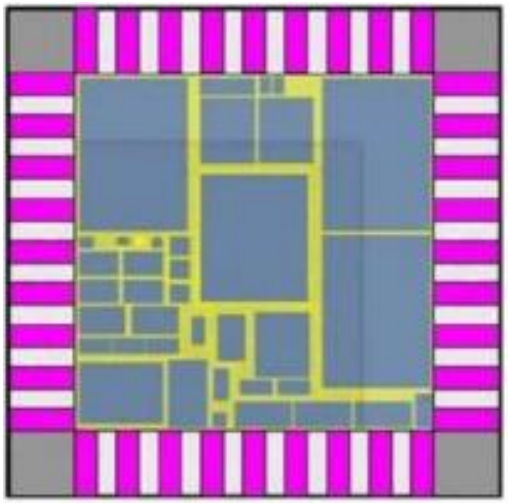
- Core power pads are used to provide power for core
- Consider core power when determining the number of core power pads
- Number of core power pads = (Power / VDD) / Max Current of Pads
- Core power pads should be evenly distributed in 4 sides
  - At least 4 core power pads

Example of Determining # of Core Power Pads

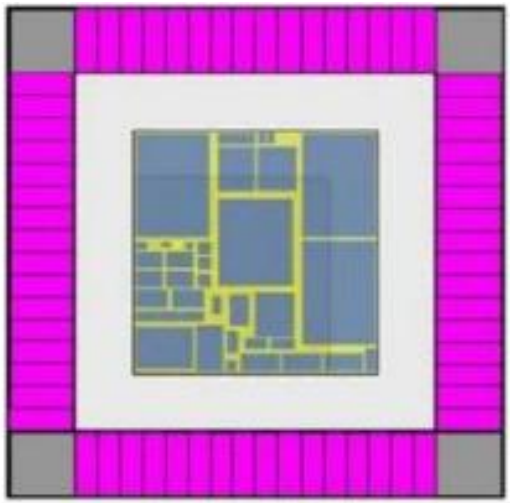
- If the core power of a chip is 32 mW, core VDD = 0.8 V
  - Core VDD pad max provided current = 38.25 mA
  - Core VSS pad max provided current = 91.87 mA
    - Number of core VDD pad = (32 / 0.8) / 38.25 = 1.05 -> 2
    - Number of core VSS pad = (32 / 0.8) / 91.87 = 0.44 -> 1
    - For evenly distributed, both number of core VDD and VSS pads = 4

Tips of IO Pads Planning

- Place more power pads if there are remaining spaces
  - Package: 128/144/160/208… pins
- Place power pads between the signal pads
- Isolate sensitive input pads (ex: clock) from other switching pads with power pads

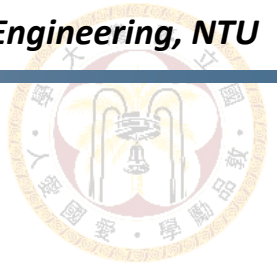


Core Limited



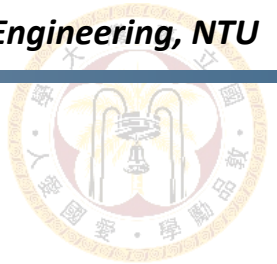
Pad Limited

Source: https://www.design-reuse.com



Outline

- Introduction
- Data Preparation
- Automatic Place & Route – Innovus
  - Design Import
  - Floorplan Specification
  - Power Planning



Getting Started for Innovus

- Source the Innovus environment (at command line)

```
source /usr/cad/innovus/CIC/license.cshrc
source /usr/cad/innovus/CIC/innovus.cshrc
```

- Invoke Innovus (at command line)

```
innovus
```

- Do not run in background mode (no &)
  - The terminal become the interface of command input while running Innovus

Innovus Tips

- The executed command is saved in innovus.cmd#
- The executed log file is saved in innovus.log#

- Restart from each step by freeing/reloading design
  - Save design

```
Innovus #> saveDesign xxx
```

  - Free Design

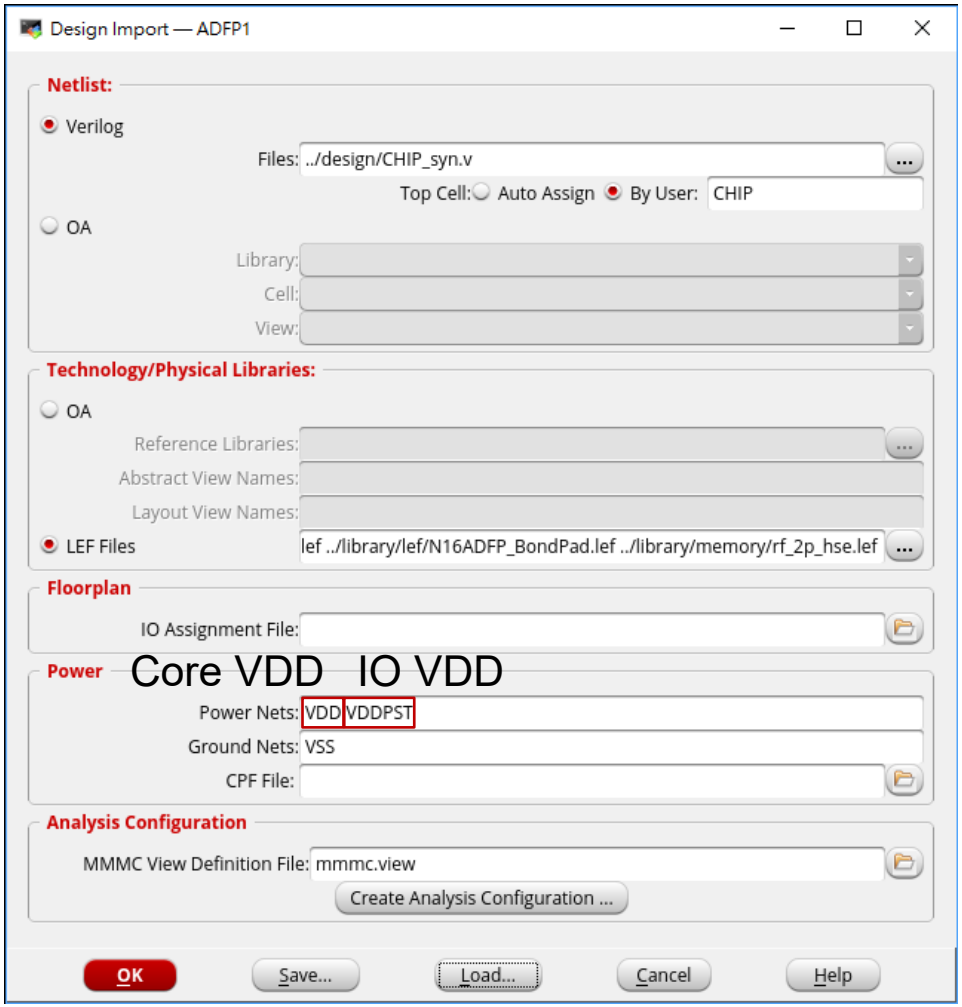
```
Innovus #> freeDesign
```

  - Restore Design

```
Innovus #> restoreDesign xxx.dat CHIP
```

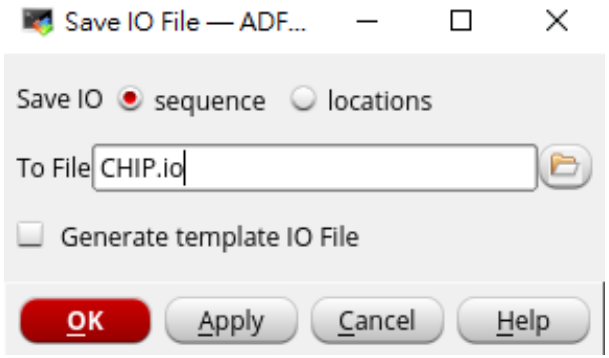
Import Design

- *File -> Design Import*
- **Import LEF in the order:**
  - Technology first
  - Geometry lef for cell/block
  - Antenna lef for cell/block (no antenna cell in ADFP)
- **IO Assignment File**
- **MMMC View Definition File**



Generate I/O Assignment File

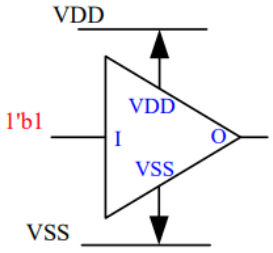
- Step 1: Import Design
- Step 2: Get an I/O assignment file template
  - ***Design -> Save -> I/O File***



- Step 3: Modify I/O assignment file
  - Adjust order
  - Add IO power, core power and corner pads
- Step 4: Reload I/O assignment file

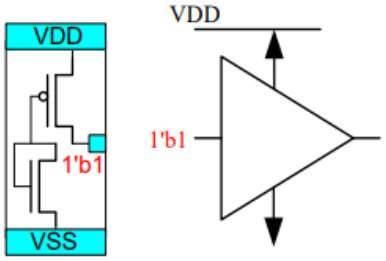
Global Net Connection

- ***Power -> Connections Global Nets***
  - VDD: core VDD
  - VDDPST: IO VDD
  - VDDPE, VDDCE, VSSE: SRAM power pin name
  - VPP: standard cell pwell tap
  - VBB: standard cell nwell tap

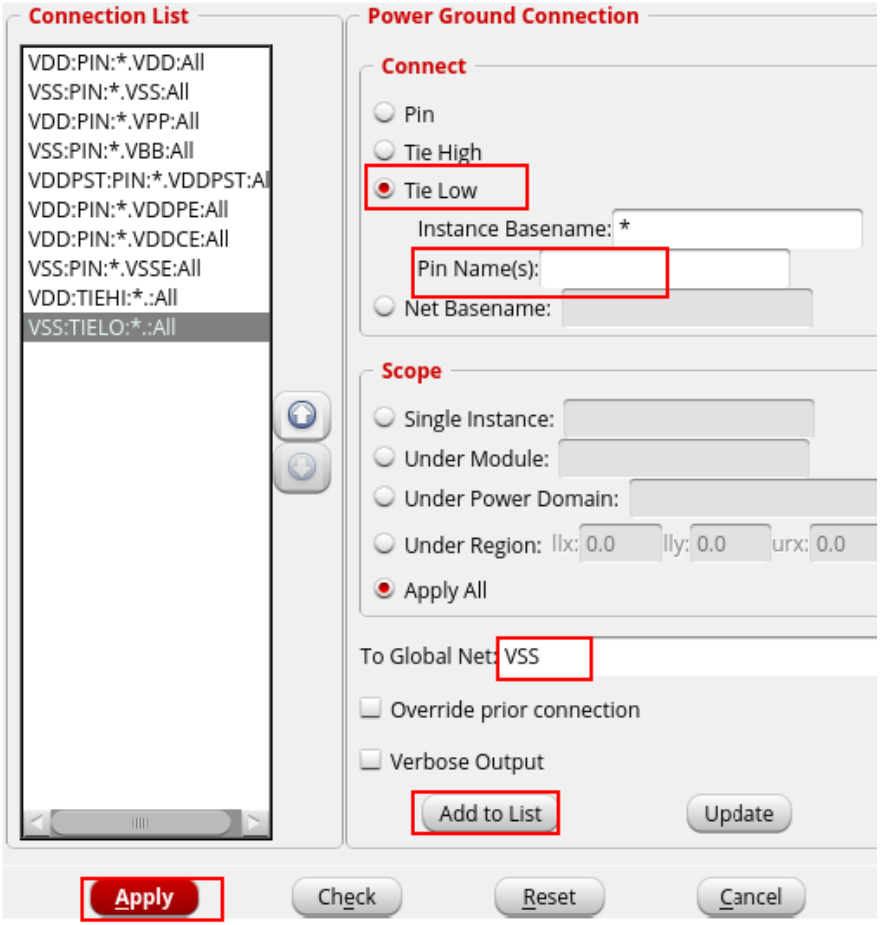


tie high net

INV inv1(.I(1'b1), .O(o));

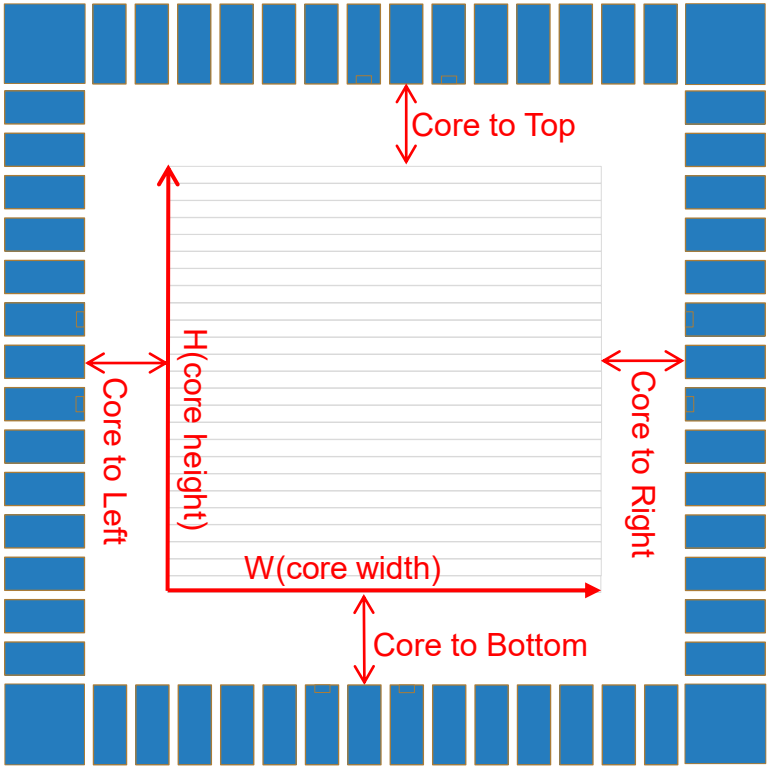


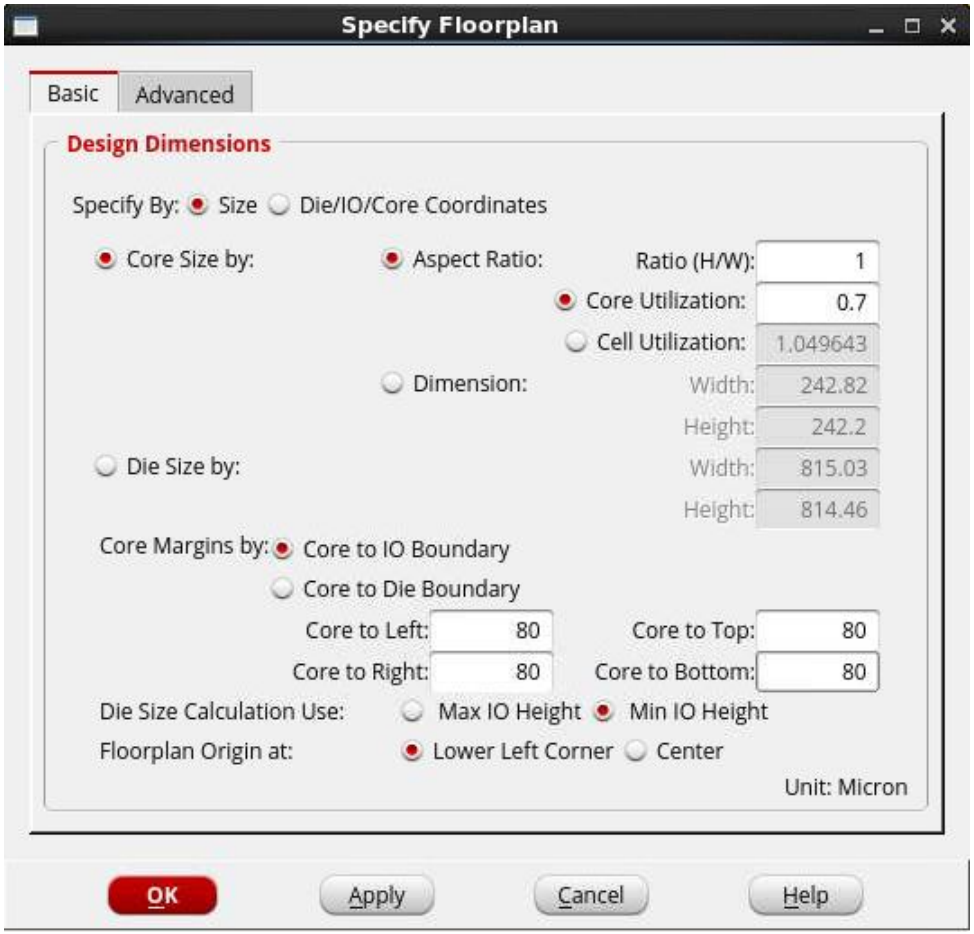
tie cell added after placement



Specify Floorplan

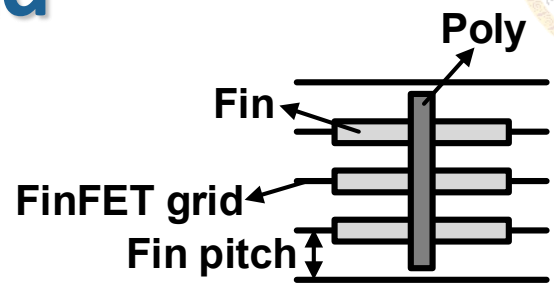
- *Floorplan -> Specify Floorplan*
  - **Aspect ratio**
  - **Utilization** (init. usually 0.7)

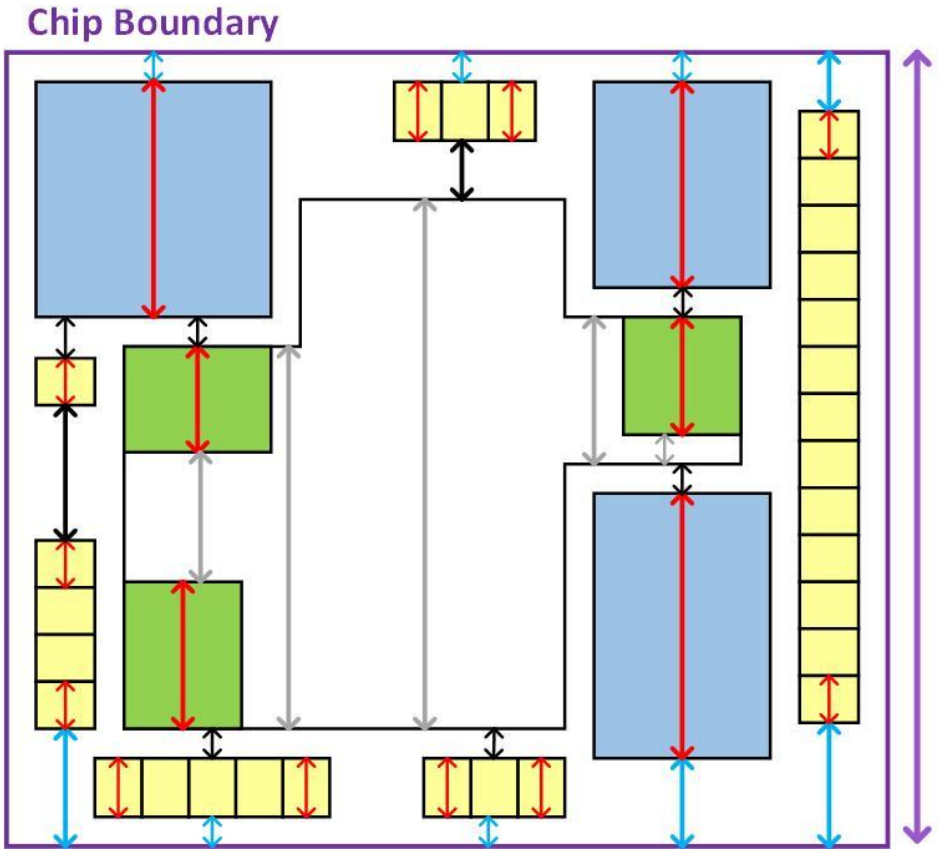




FinFET Grid

- Everything should be put on grid





Analog IP

Digital Block

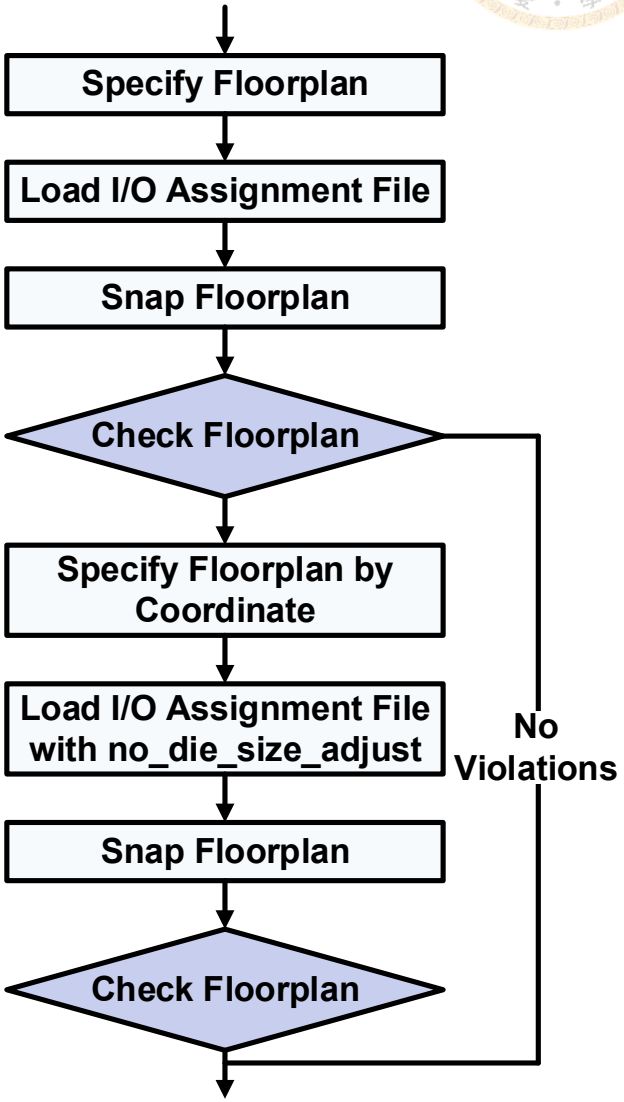
Digital I/O

Memory IP

- Chip boundary height = Even integer of ${Fin Pitch}
- Macro cell height = Integer of ${Fin Pitch}
- Digital block cell height = Integer of ${Fin Pitch}
- Cell boundary spacing = Integer of ${Fin Pitch}
- Chip boundary to cell spacing = Integer of ${Fin Pitch}

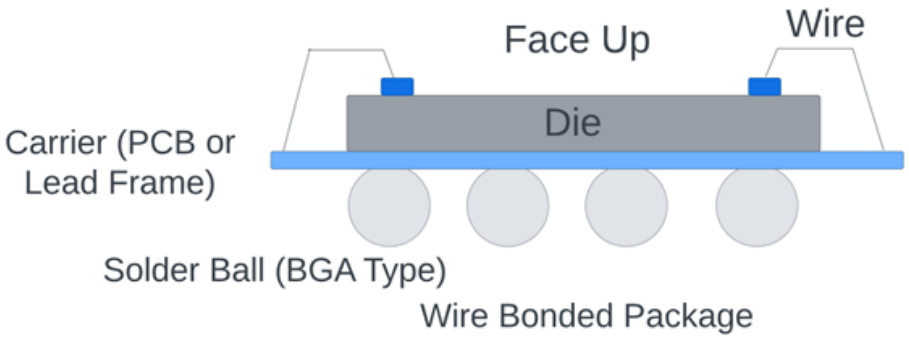
FinFET Grid

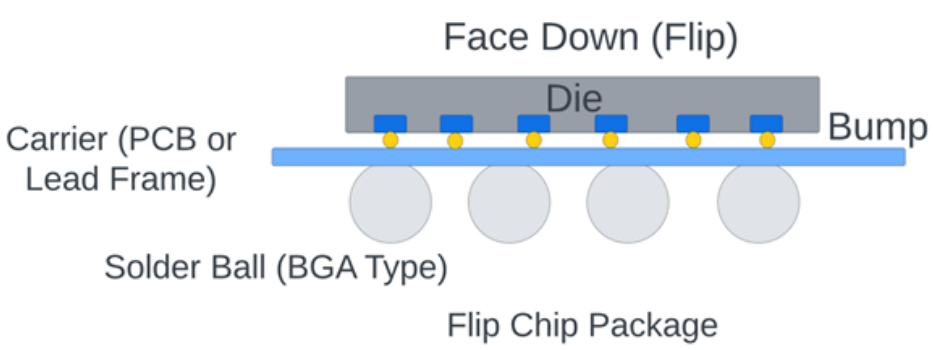
- Check the floorplan is on grid or not
  - `Innovus #> checkFPlan`
- If there is off grid error
  - `Innovus #> snapFPlan -all`
- Flow of design import
  - Specify floorplan
  - Load I/O assignment file
  - Snap floorplan
  - Check floorplan



Flip Chip

- An advanced packaging technology whereby the chip is flipped upside-down and connected directly to the substrate using small solder bumps
- Bumps serve as electrical and mechanical connections between the chip and the substrate
- Higher I/O density: allow for more connections in a smaller area

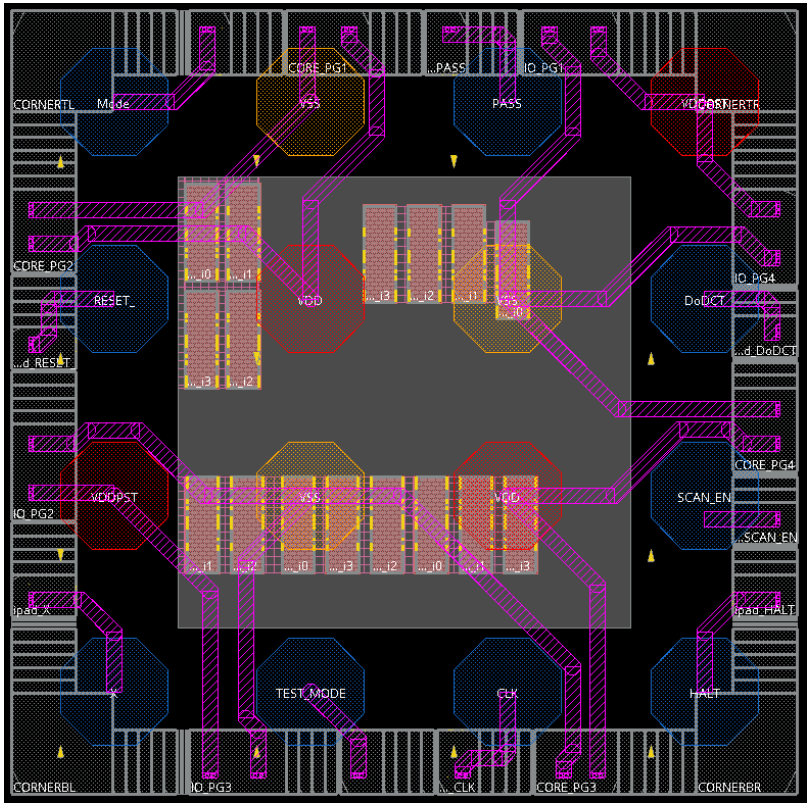


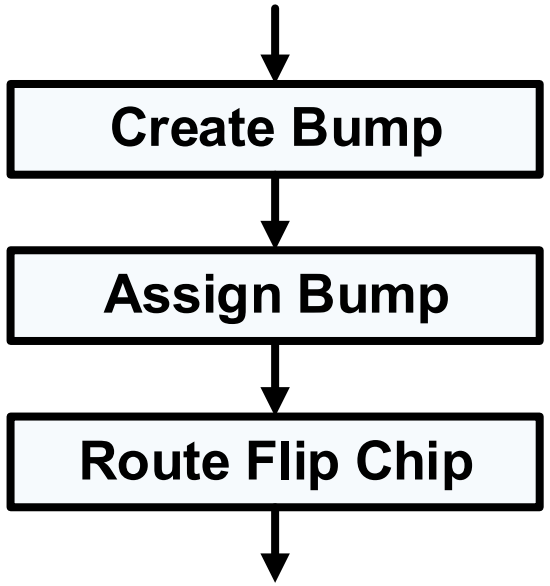


Source: https://anysilicon.com

Flip Chip Flow

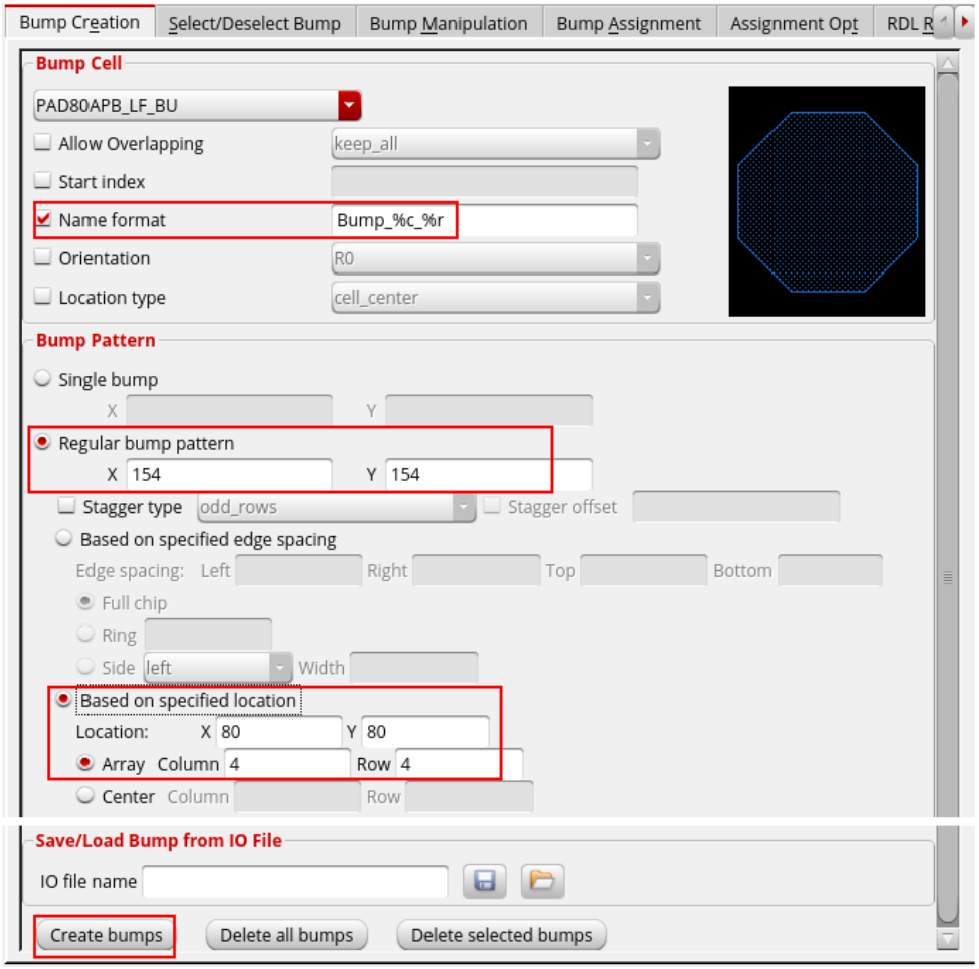
- Create bumps on the floorplan
- Assign bumps to the signal that they would connect to
- Trial connect between pads and bumps

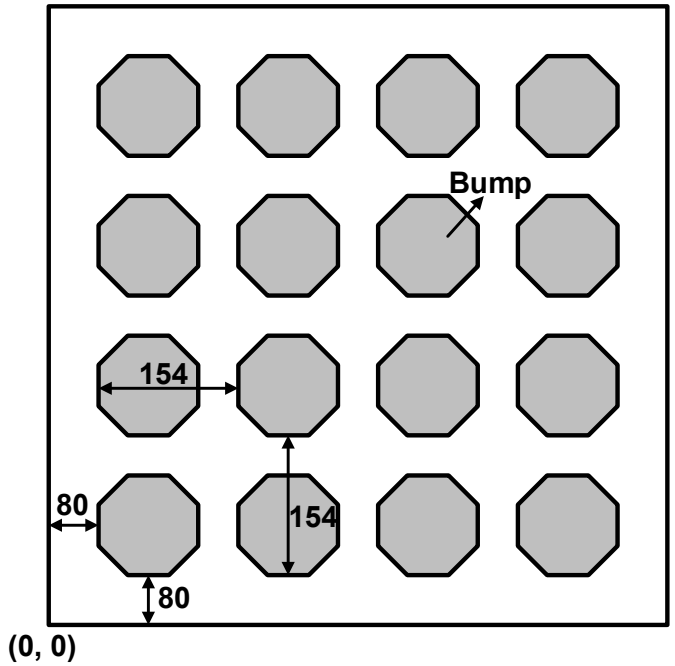




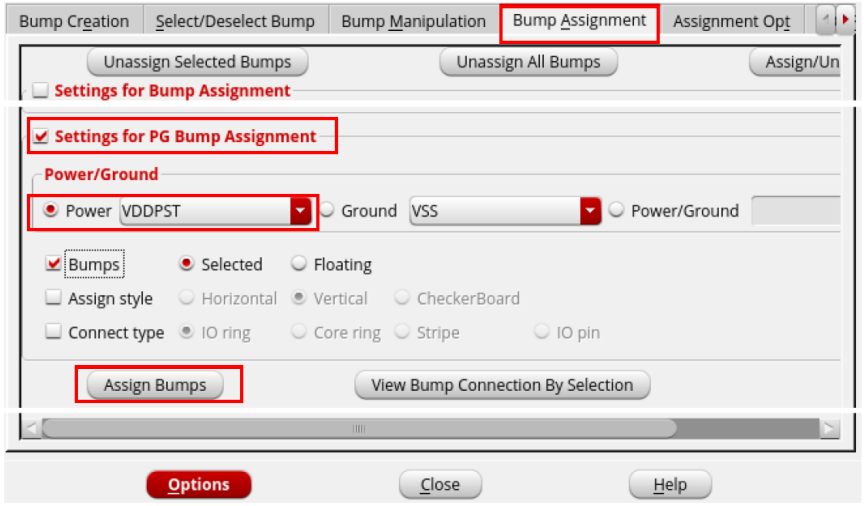
Create Bump

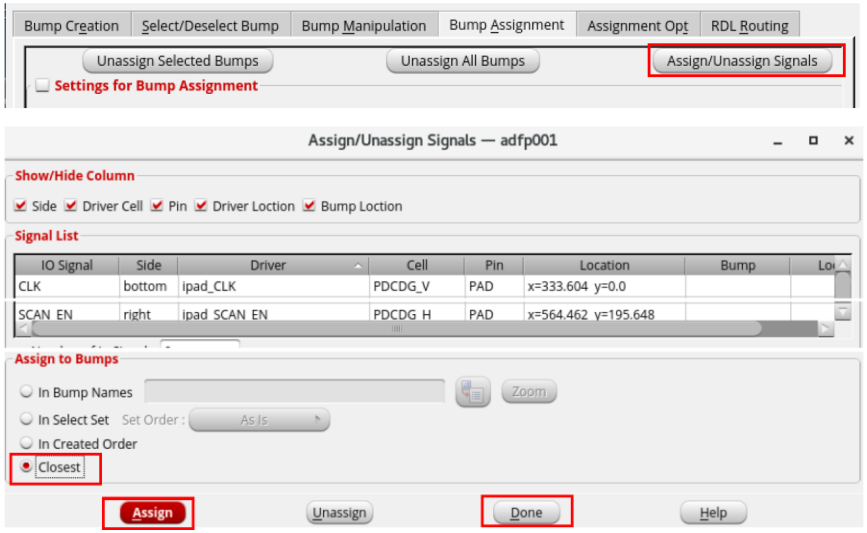
- *Tools -> Flip Chip*



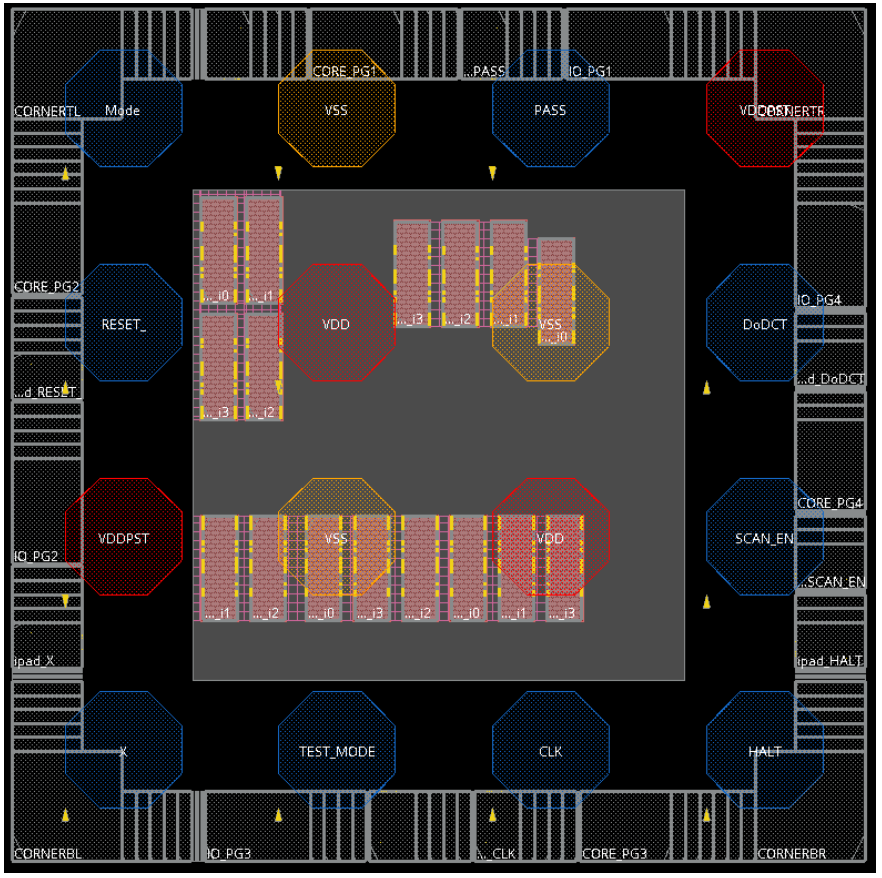


Assign Bump



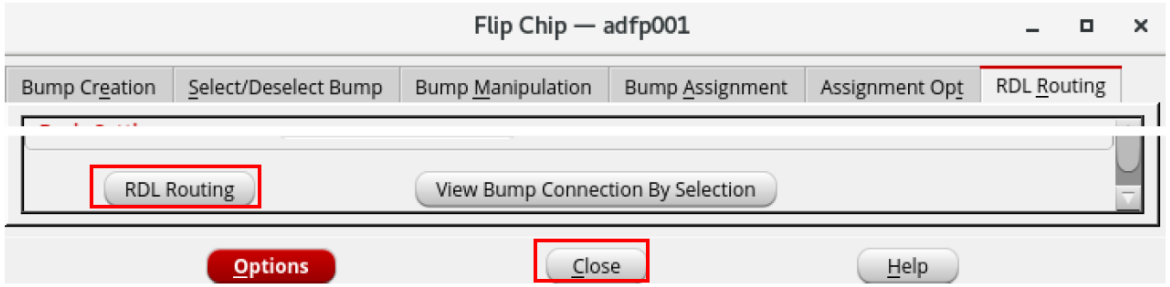


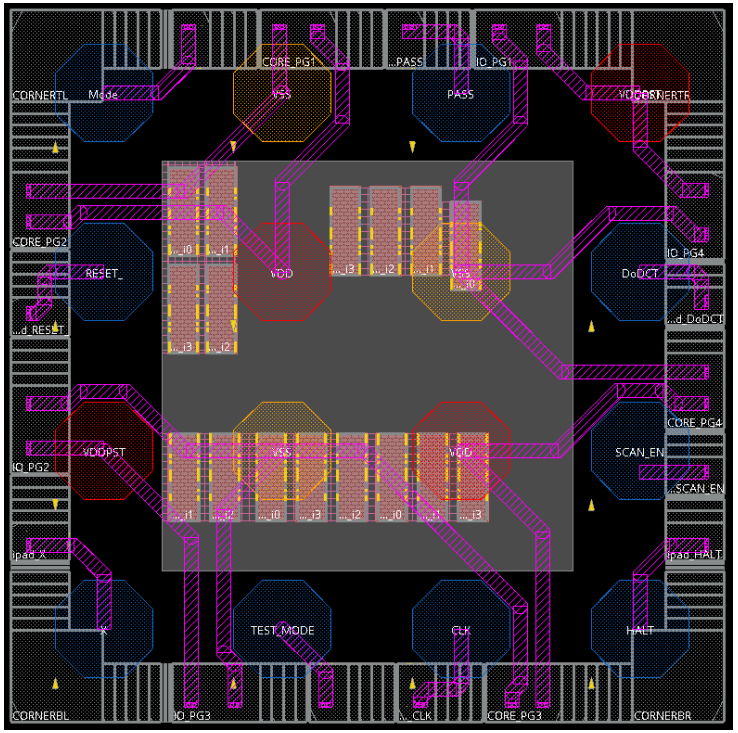
- Can connect multiple power pads to a single power bump



RDL Prerouting

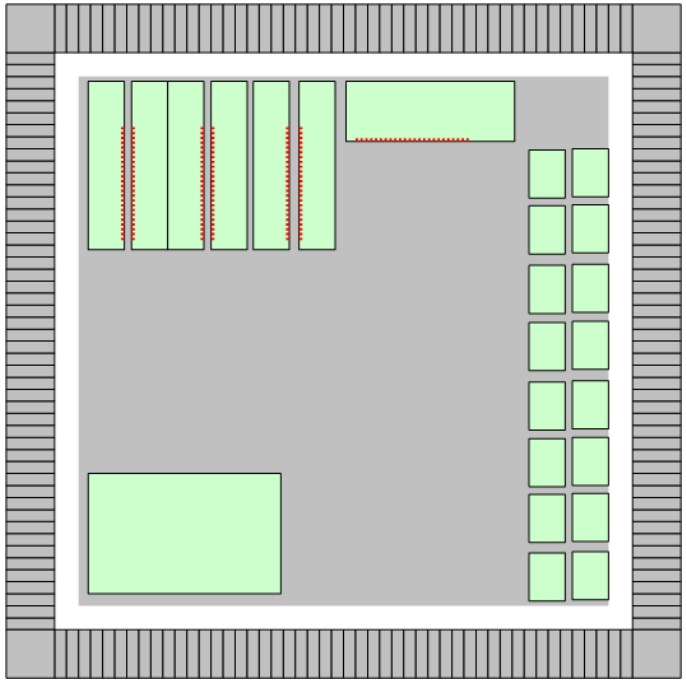
- RDL preroute in this stage (after floorplan)
  - Check RDL route can success under this I/O placement
- Delete bumps and routing after RDL route success
- Redo flip chip flow after adding core filler





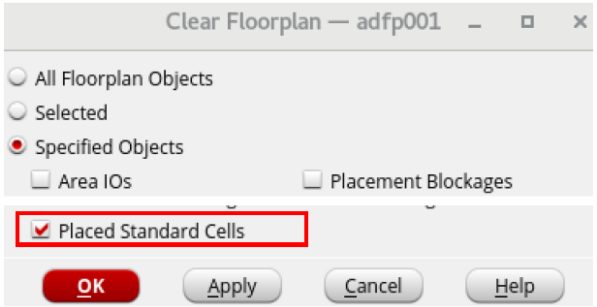
Floor Planning with Hard Macro

- Tips for macro placement
  - Place macros around chip periphery
  - Consider connections to fixed cells when placing macros
  - Orient macros to minimize distance between pins
  - Reserve space for power grid and signal routing and possible buffer insertion
  - Keep out standard cells with block halo
  - Keep random areas continuous to avoid detaches areas
  - Keep edges of macros aligned if possible

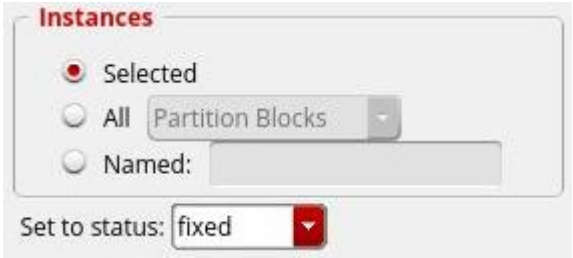


Place Macro Block

- Automatic generate a quick, initial floorplan
  - ***Floorplan -> Generate Floorplan -> Place Macros***
  - Innovus also place standard cells when auto place macros
  - Clear standard cells after auto place macros
    - ***Floorplan -> Clear Floorplan***

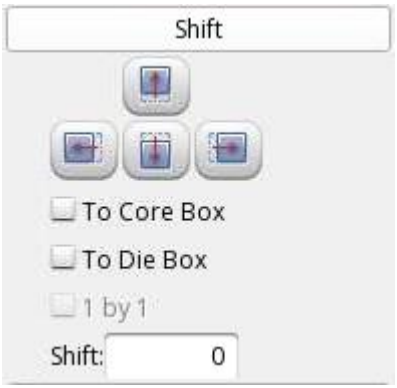


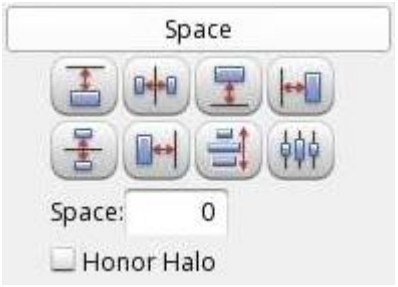
-  Move/Resize/Reshape floorplan object
- Fix block placement status
  - ***Floorplan -> Edit Floorplan -> Set Instance Placement Status***



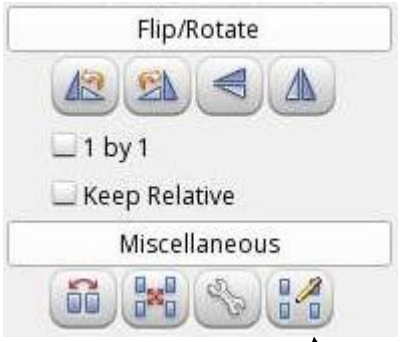
Place Macro Block

- Manually place macros for better macros placement result
- Use floorplan toolbox
  - ***Floorplan -> Floorplan Toolbox***





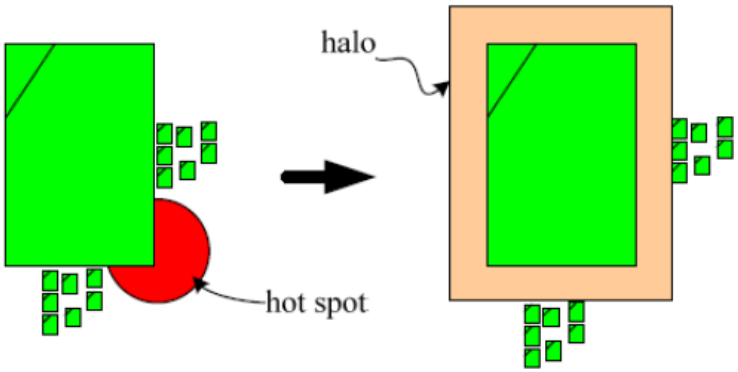


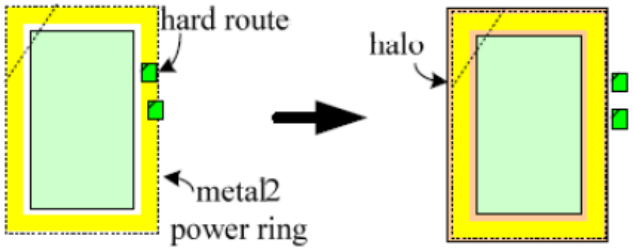


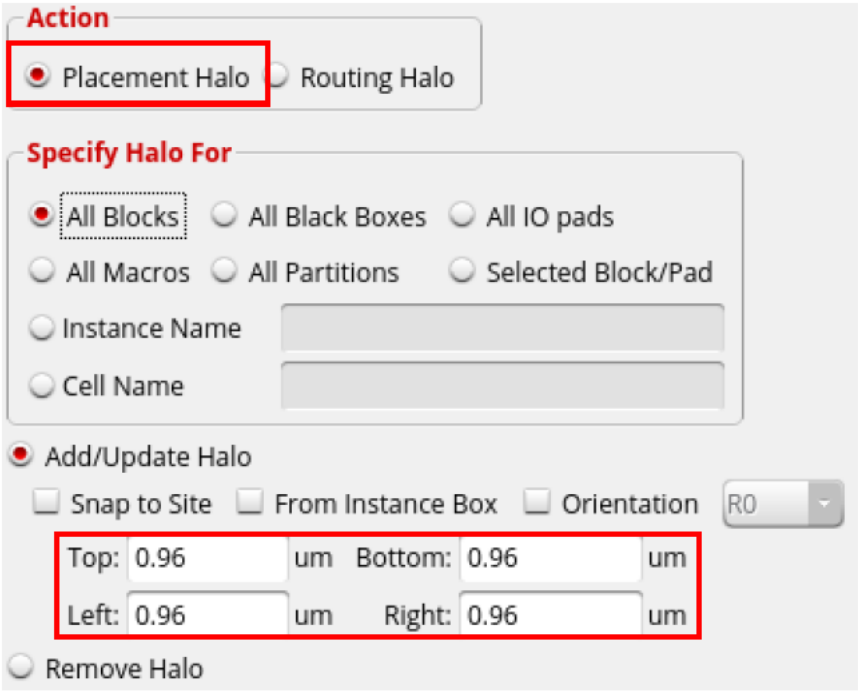
set placement status

Add Placement Halo to Block

- ***Floorplan -> Edit Floorplan -> Edit Halos***
  - Prevent the placement of blocks and standard cells in order to reduce congestion around a block

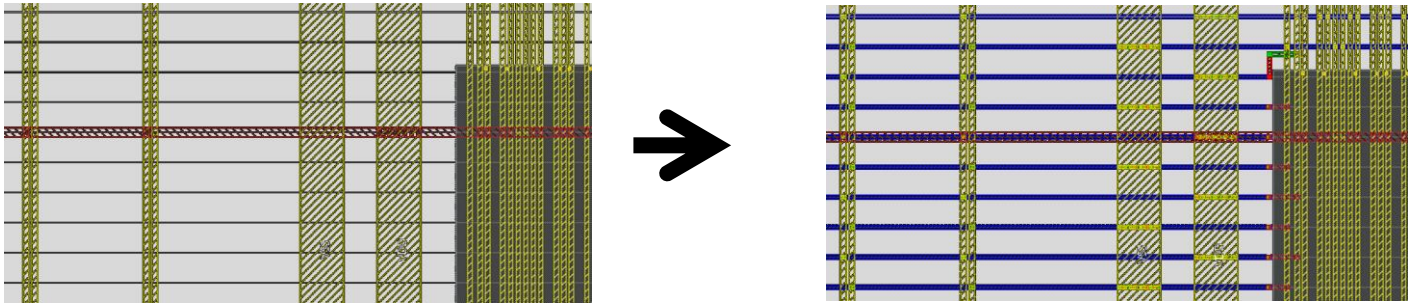




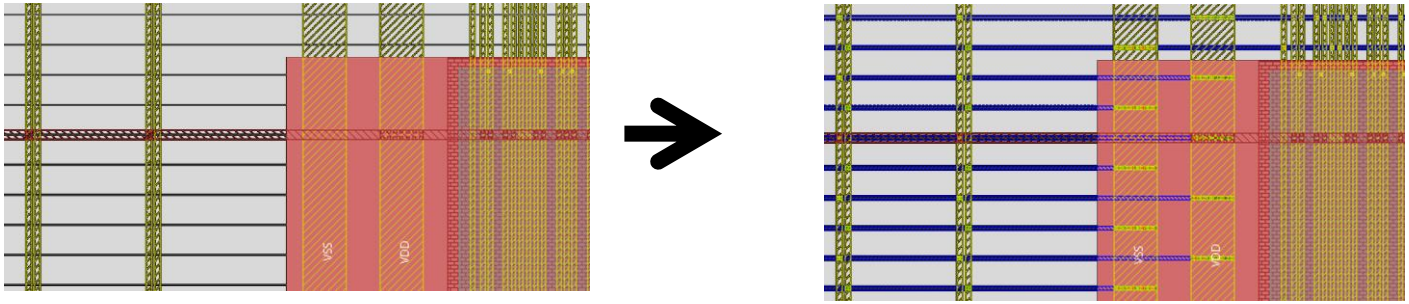


Placement Halo Affect Followpin

Without Halo

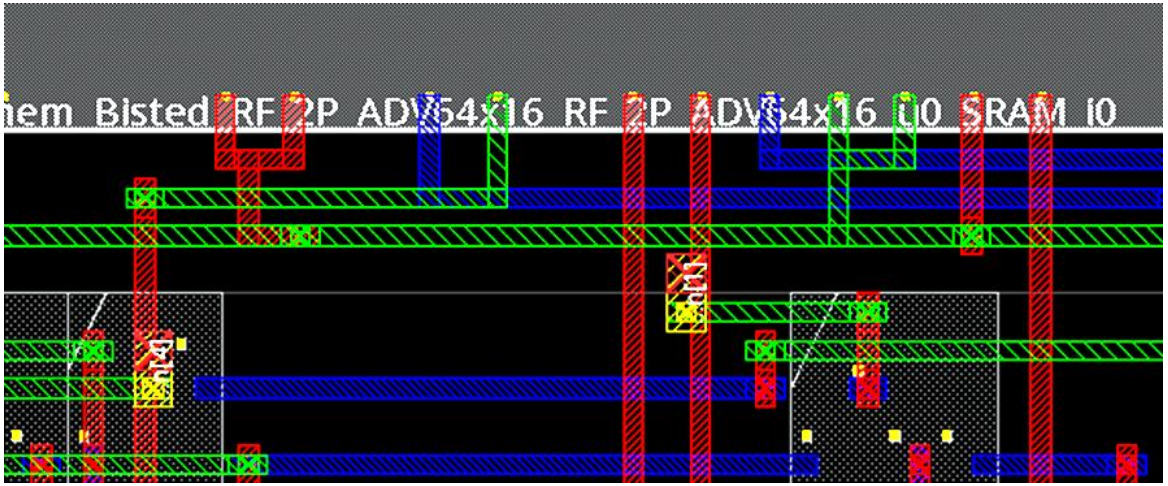


With Halo

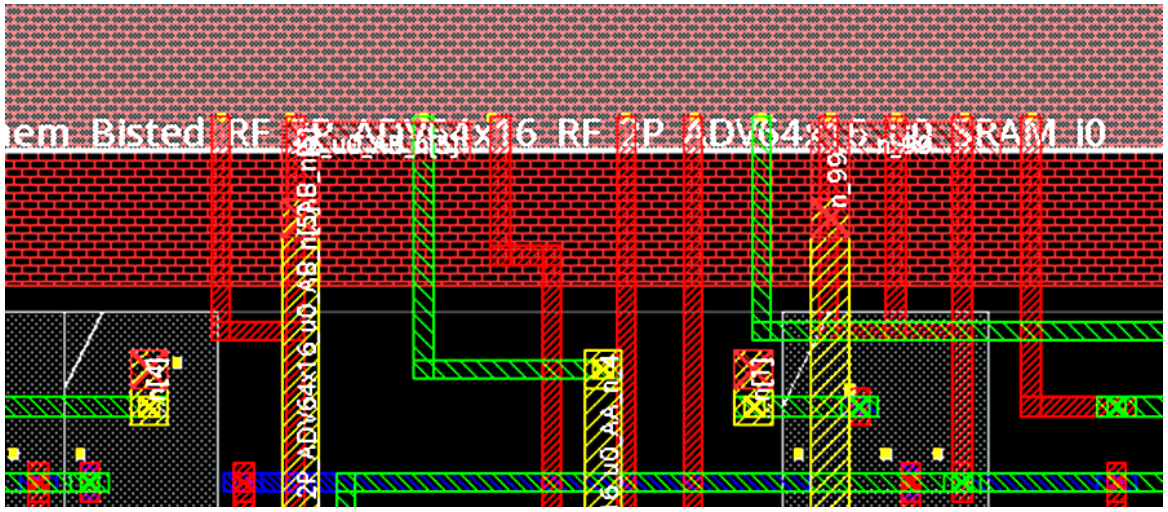


Add Routing Halo to Block (Not Necessary)

- Reduce the possibility of long wire routing within



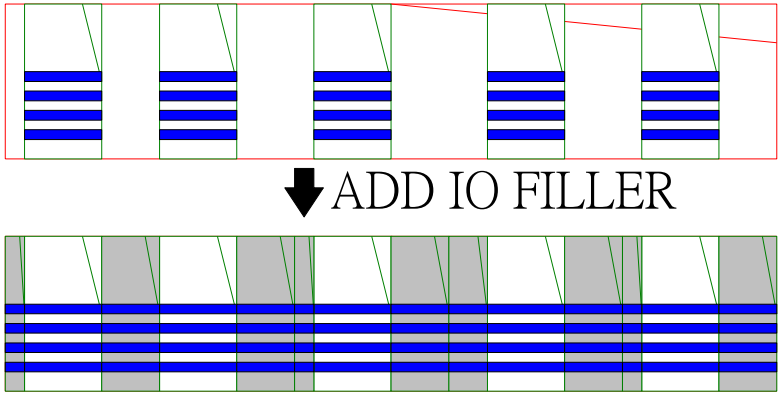
without routing halo

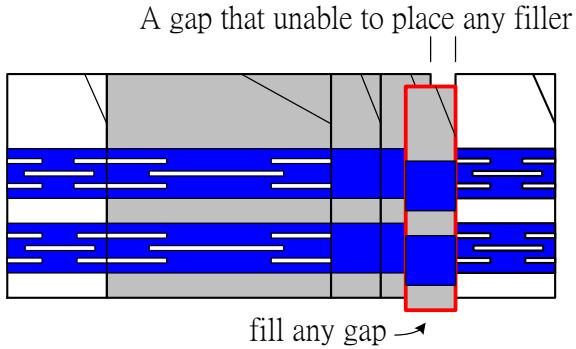


with routing halo

Connect Pad Power Ring

- Connect IO pad power bus by inserting IO filler



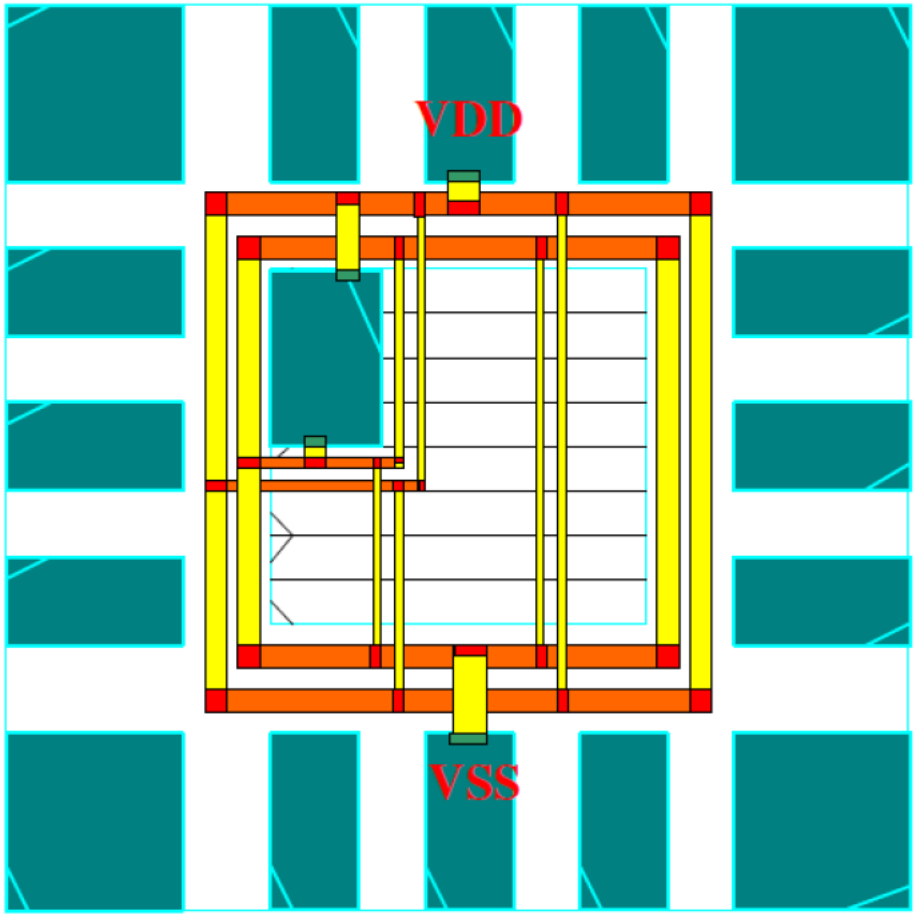


- Add from wider filler to narrower filler
  - Directly source add_io_fillers.tcl

```
addIoFiller -cell PFILLER10080 -prefix IOFILLER -logic -deriveConnectivity
addIoFiller -cell PFILLER01008 -prefix IOFILLER -logic -deriveConnectivity
addIoFiller -cell PFILLER00048 -prefix IOFILLER -logic -deriveConnectivity
addIoFiller -cell PFILLER00001 -prefix IOFILLER -logic -deriveConnectivity -fillAnyGap
```

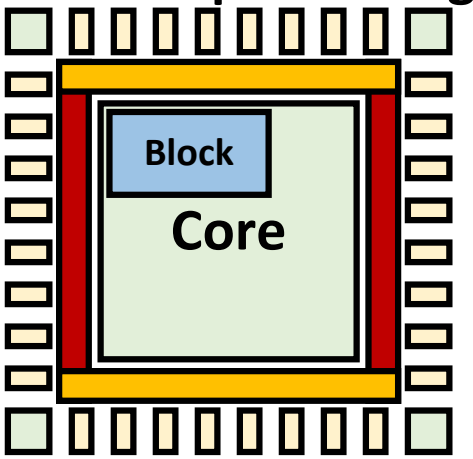
Power Planning Order

- **Concept: connect wider nets prior then narrow ones**
  - Create power ring
  - Connect pad pin
  - Create block ring
  - Connect block pin
  - Create stripe
  - Connect follow pin

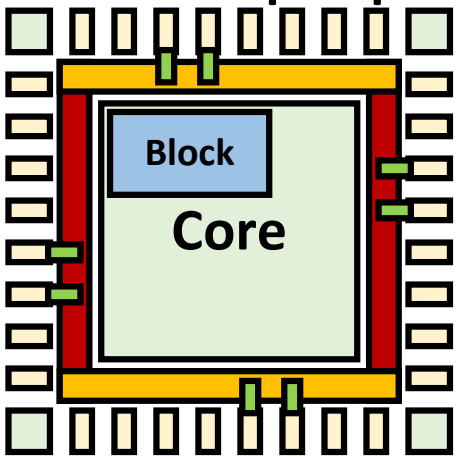


Power Planning Overview

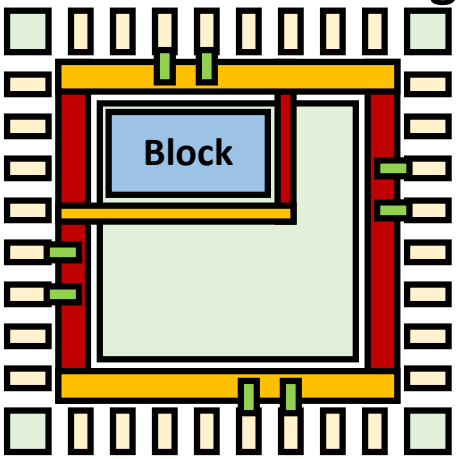
1. Create power ring



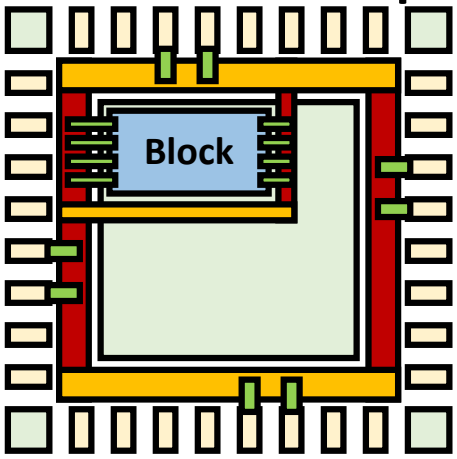
2. Connect pad pin



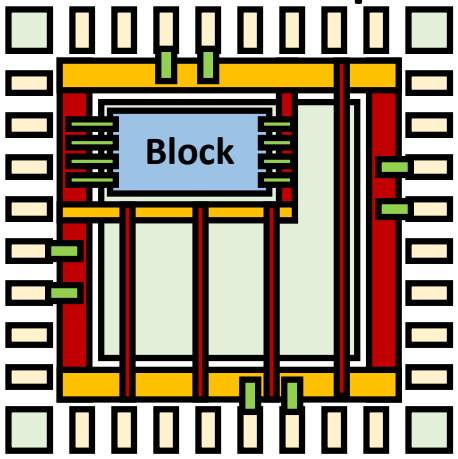
3. Create block ring



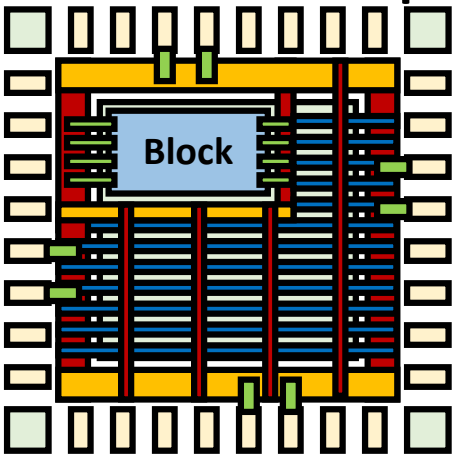
4. Connect block pin



5. Create stripe



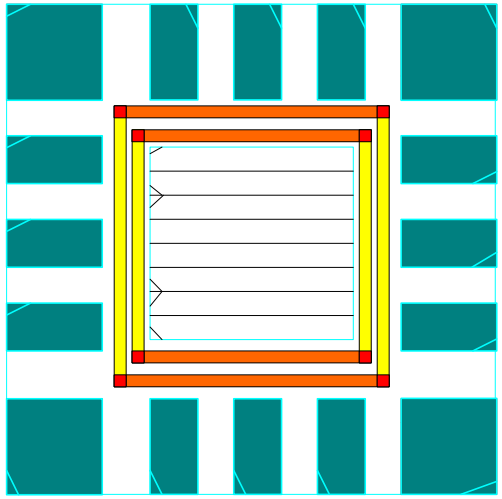
6. Connect follow pin



Power Planning: Power Ring

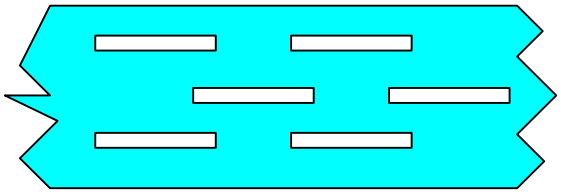
- *Power -> Power Planning -> AddRings*
- **Use wire group and interleaving to avoid slot DRC error**



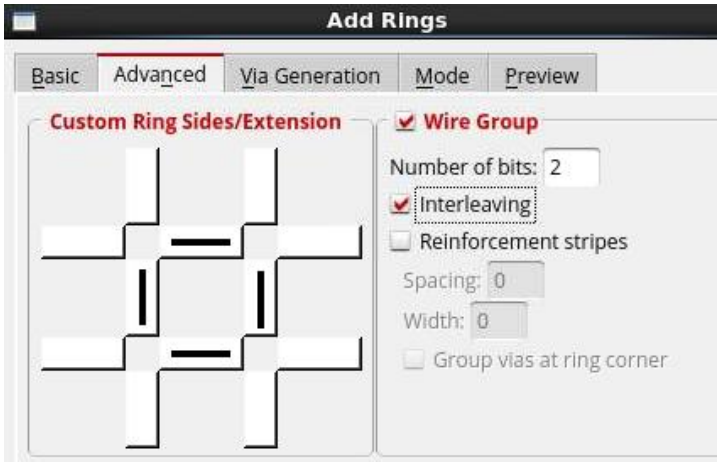


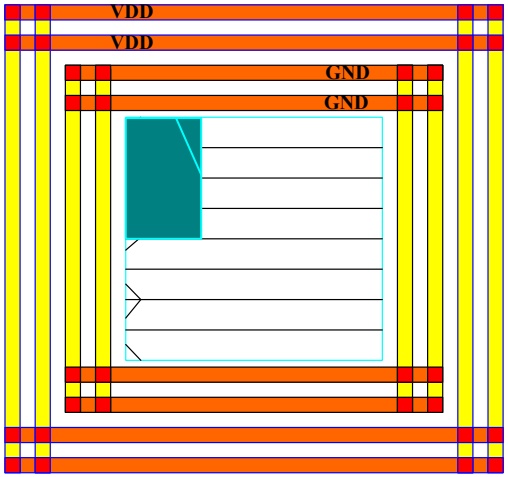
Power Planning: Power Ring

- No metal slot (to relieve thermal stress) in Innovus
  - Use wire group instead

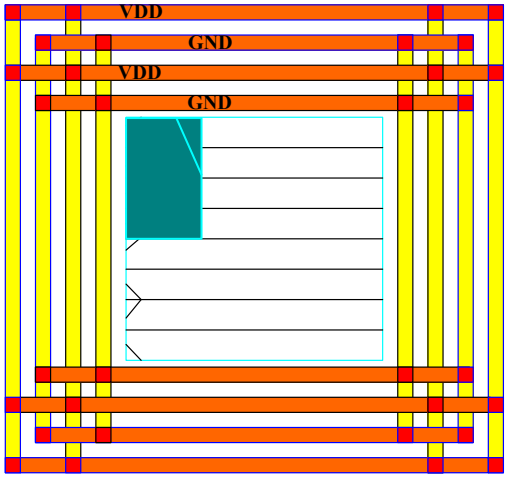


metal slot





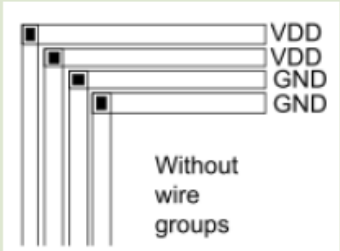
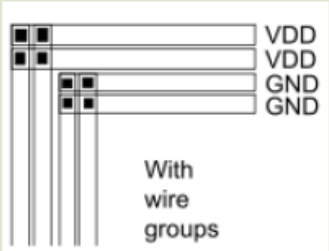
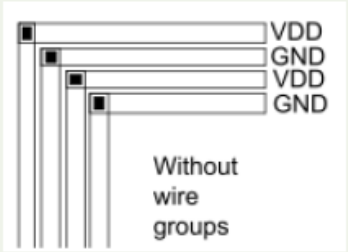
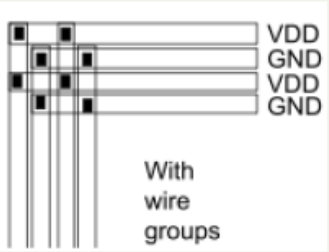
wire group



wire group & interleaving

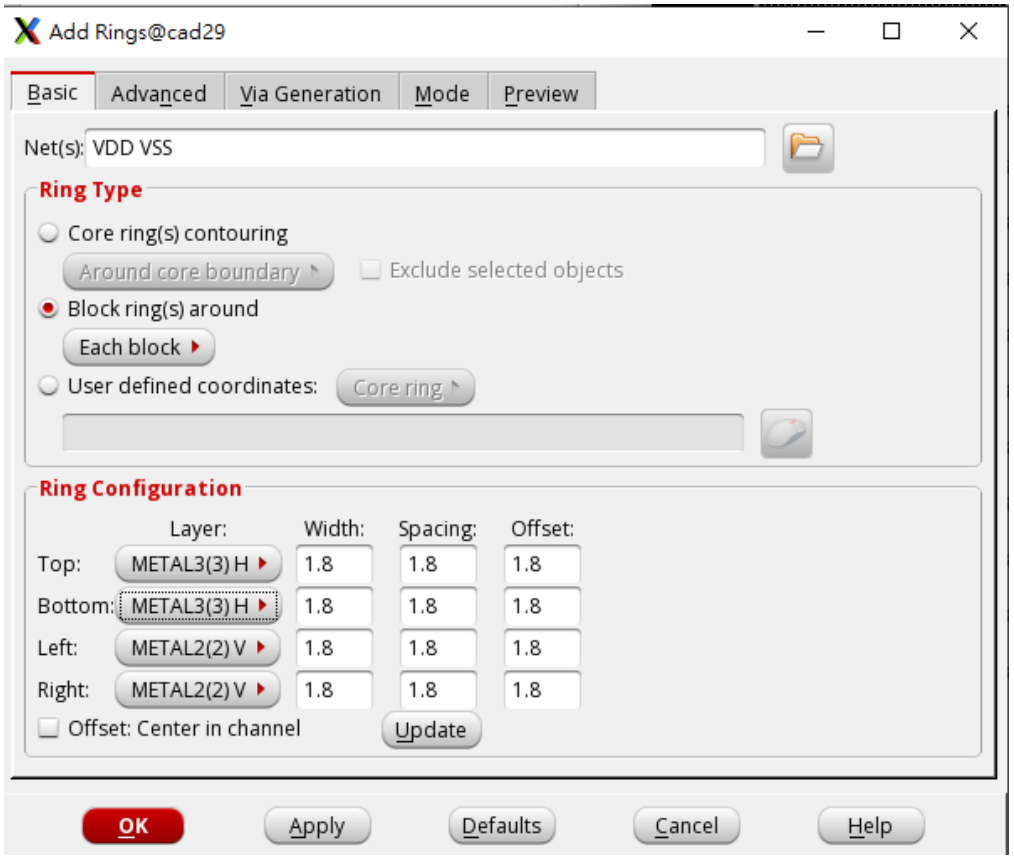
Power Planning: Power Ring

- **Supply voltage for the core evenly by power pads**
- **Interleaving and wire group**
  - To increase the coupling capacitance of power rings
  - Thus reduce the power noise

	Without Wire group	With Wire group
Without Interleaving		
Interleaving		

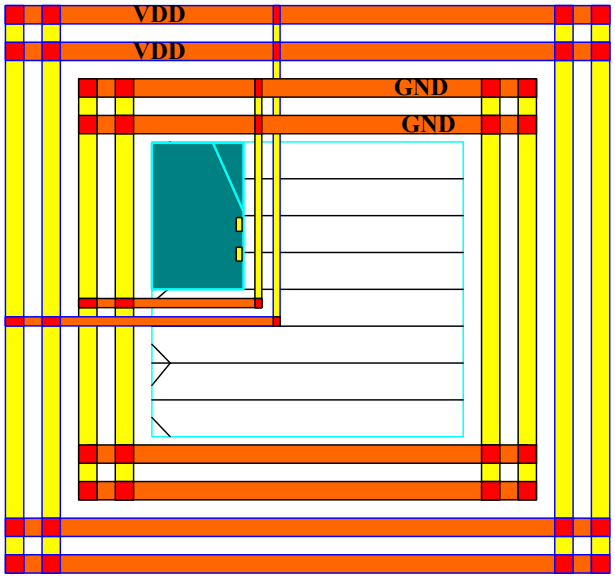
Power Planning: Block Ring

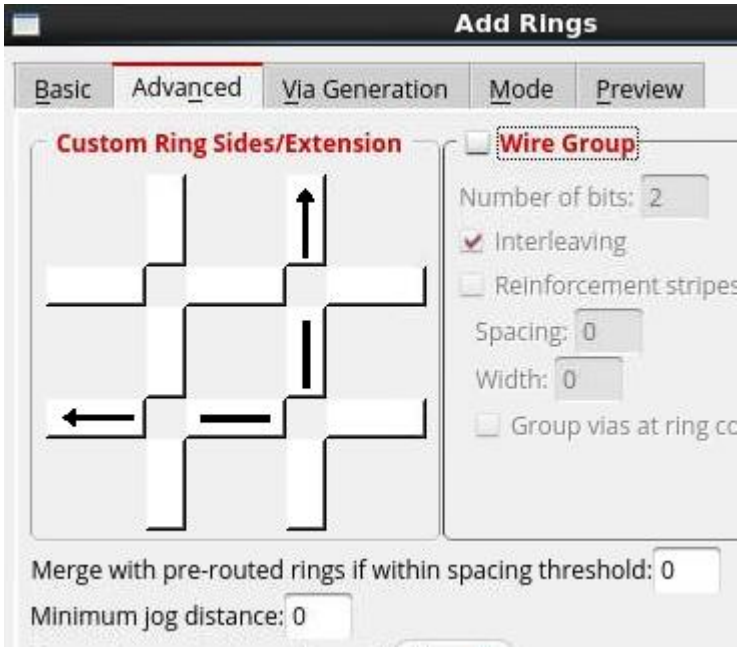
- **Supply voltage for hard macros**
- **Add block ring**
  - Specify nets
  - Block ring around
  - Specify metals
  - Width, offset
  - Extension
  - Wiregroups



Power Planning: Block Ring

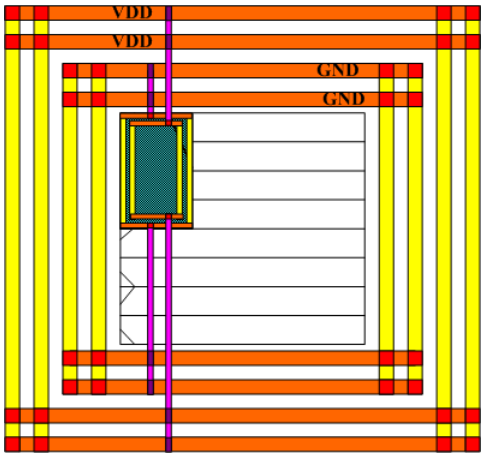
- Select the macros and choose the extension direction





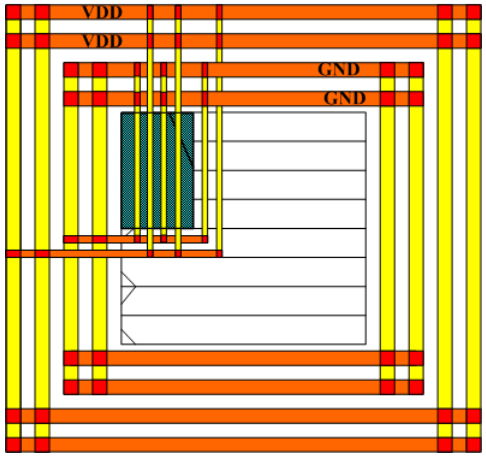
Connect Block Pin

ring type



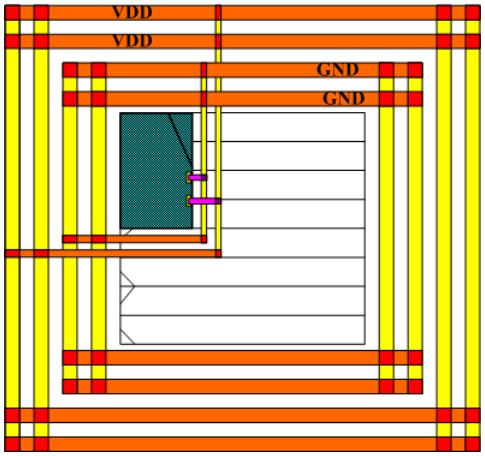
add stripes
- set to set distance

rail type (vertical)



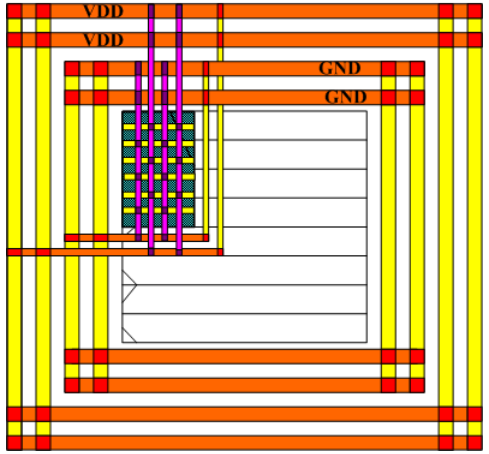
add stripes
- over p/g pins

pin type



SRoute
- block

rail type (horizontal)



add stripes
- set to set distance